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Yang et al.

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(54) **FULL WELL CAPACITY FOR IMAGE SENSOR**

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(Continued)

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H01L 27/146 (2006.01)
H10F 30/221 (2025.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10F 77/1243** (2025.01); **H10F 30/2215** (2025.01); **H10F 39/028** (2025.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 31/03042; H01L 27/14643; H01L 27/14698; H01L 31/03529; H01L 31/1035;
(Continued)

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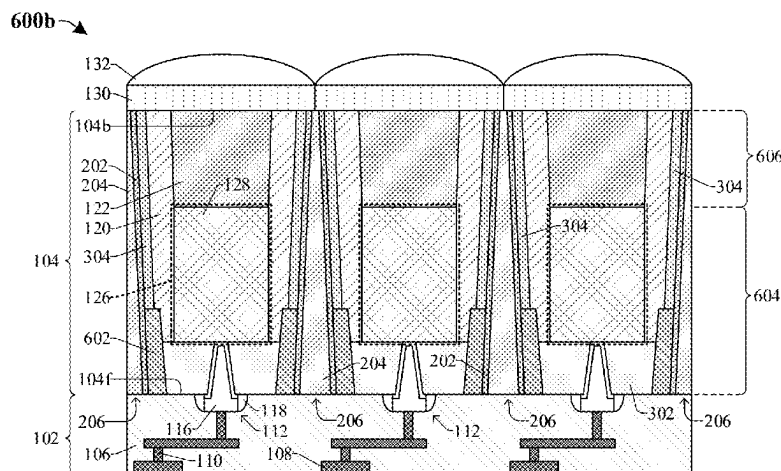
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(57) **ABSTRACT**

Various embodiments of the present disclosure are directed towards an image sensor having a photodetector disposed in a semiconductor substrate. The photodetector comprises a first doped region comprising a first dopant having a first doping type. A deep well region extends from a back-side surface of the semiconductor substrate to a top surface of the first doped region. A second doped region is disposed within the semiconductor substrate and abuts the first doped region. The second doped region and the deep well region comprise a second dopant having a second doping type opposite the first doping type. An isolation structure is disposed within the semiconductor substrate. The isolation structure extends from the back-side surface of the semiconductor substrate to a point below the back-side surface. A doped liner is disposed between the isolation structure and the second doped region. The doped liner comprises the second dopant.

20 Claims, 20 Drawing Sheets



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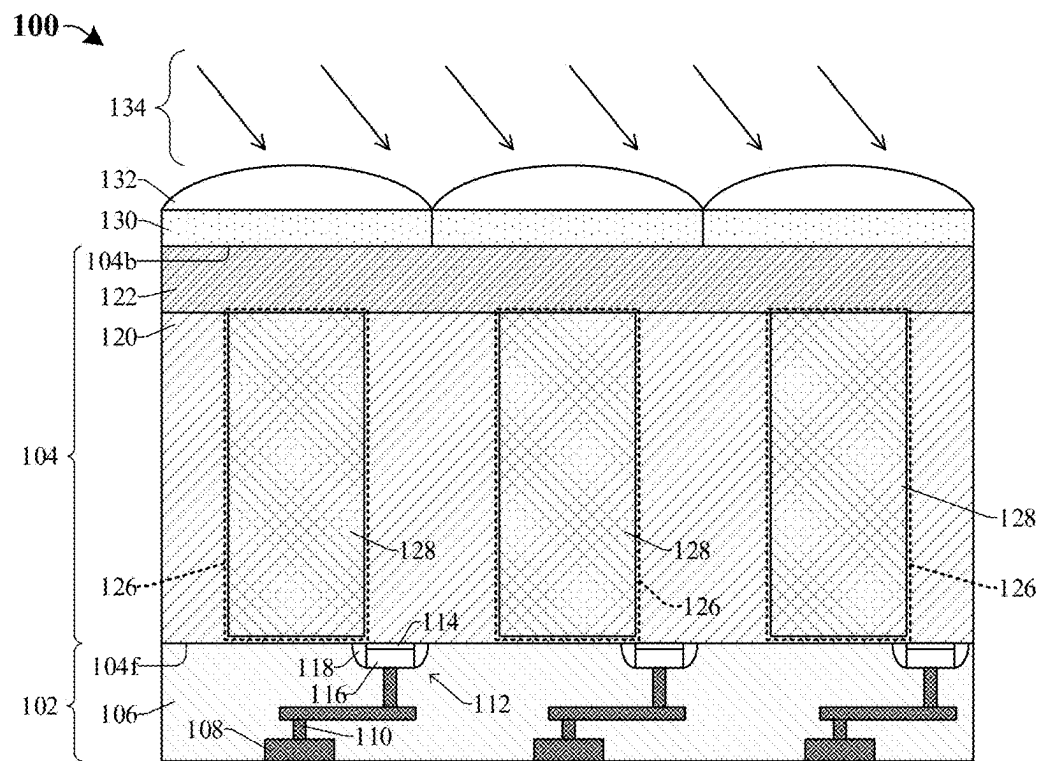


Fig. 1

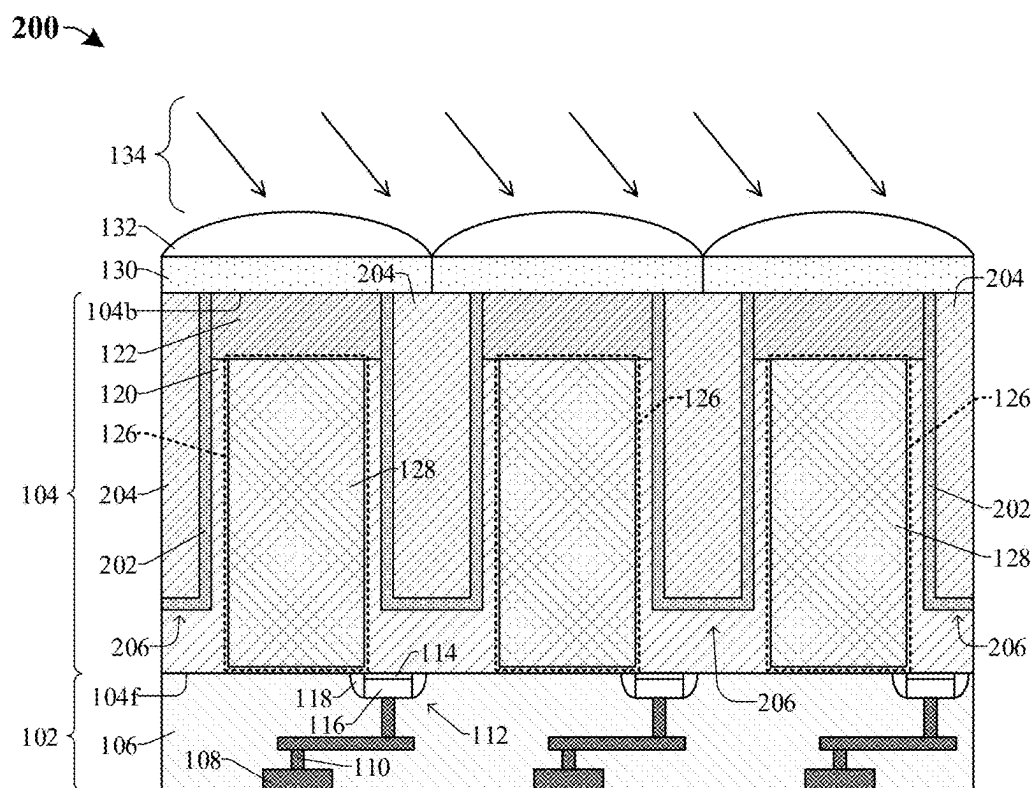


Fig. 2

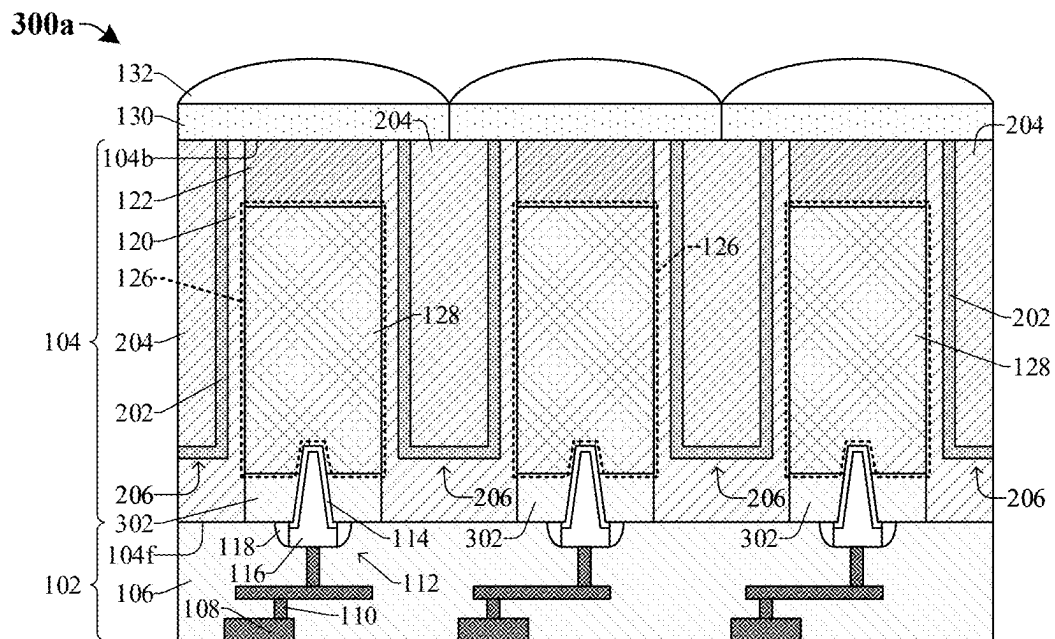


Fig. 3A

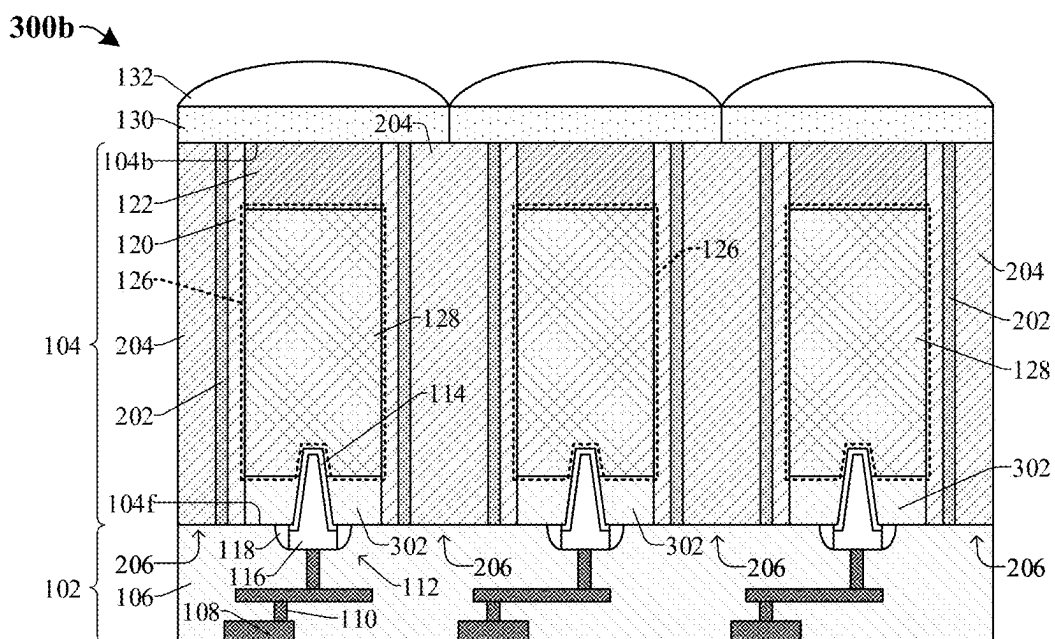


Fig. 3B

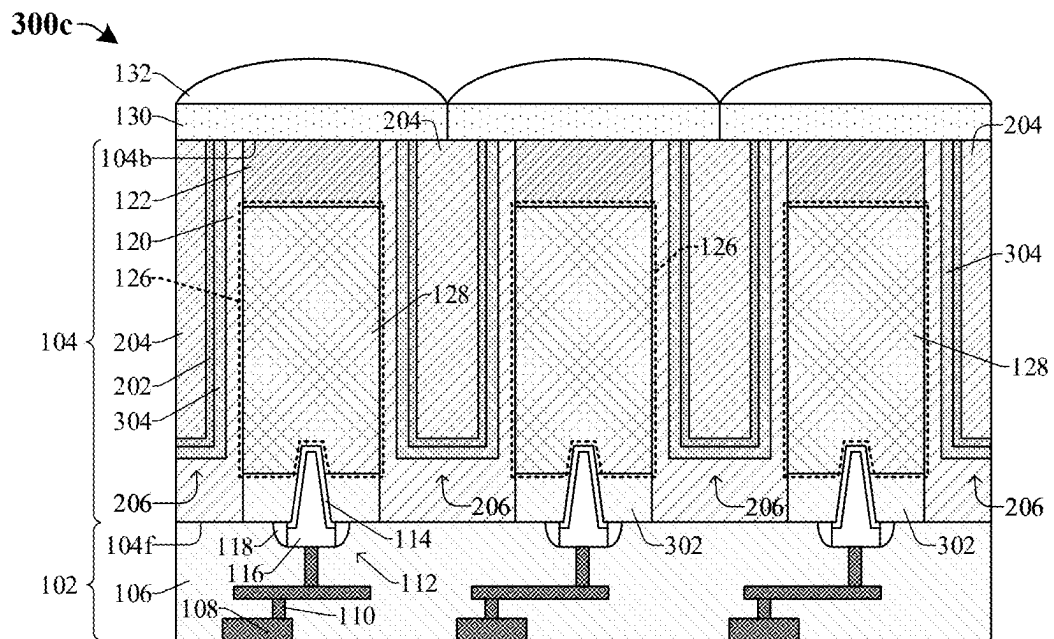


Fig. 3C

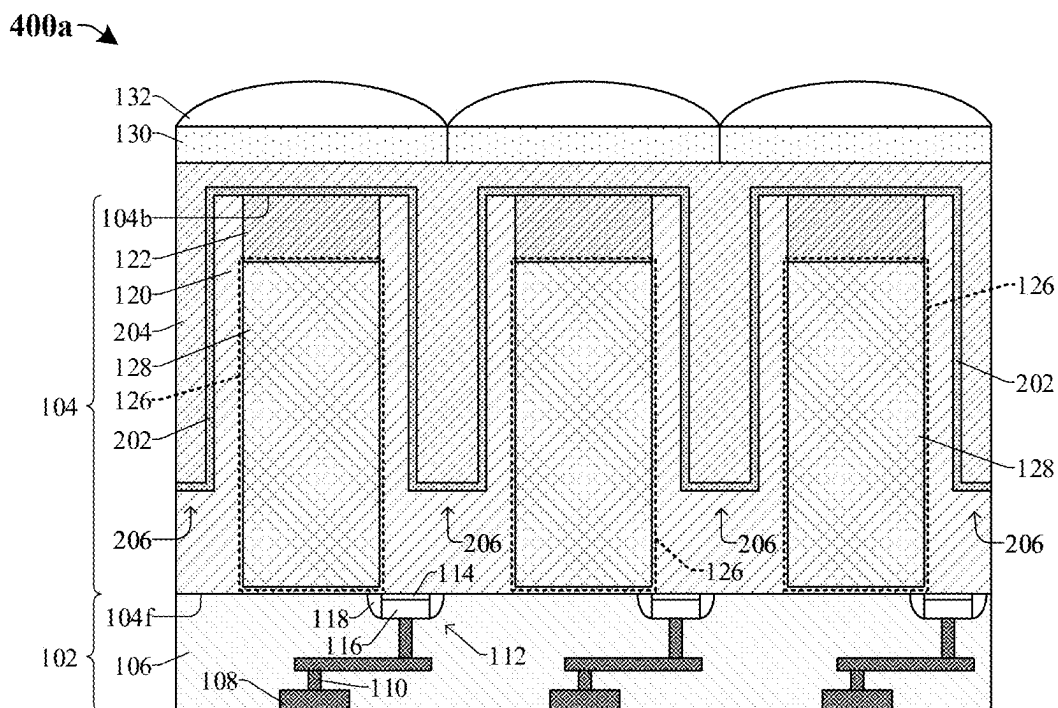


Fig. 4A

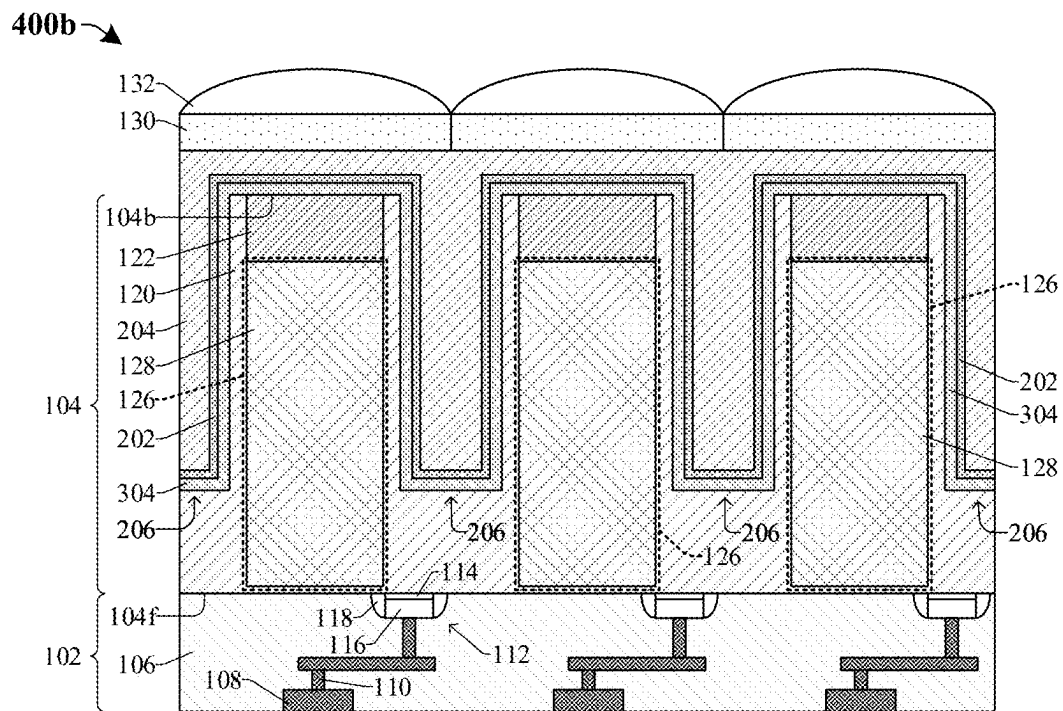


Fig. 4B

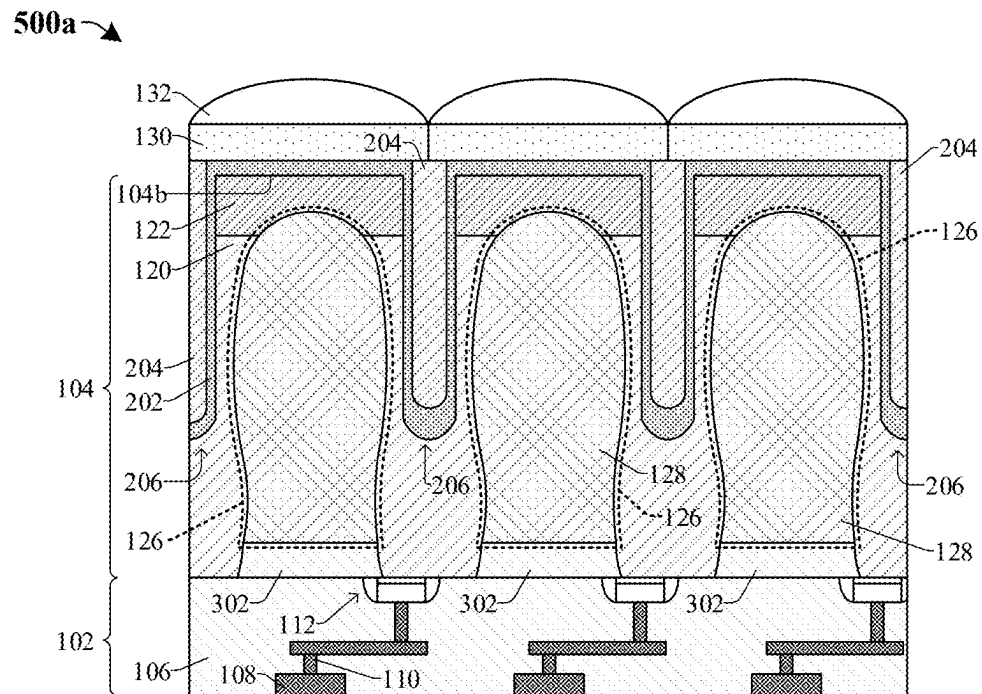


Fig. 5A

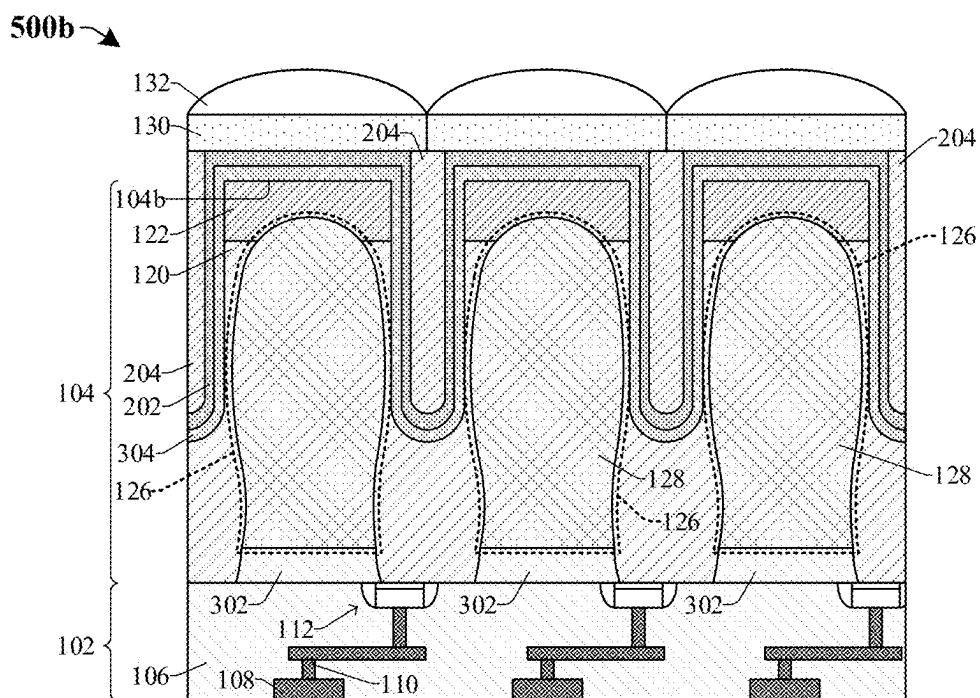


Fig. 5B

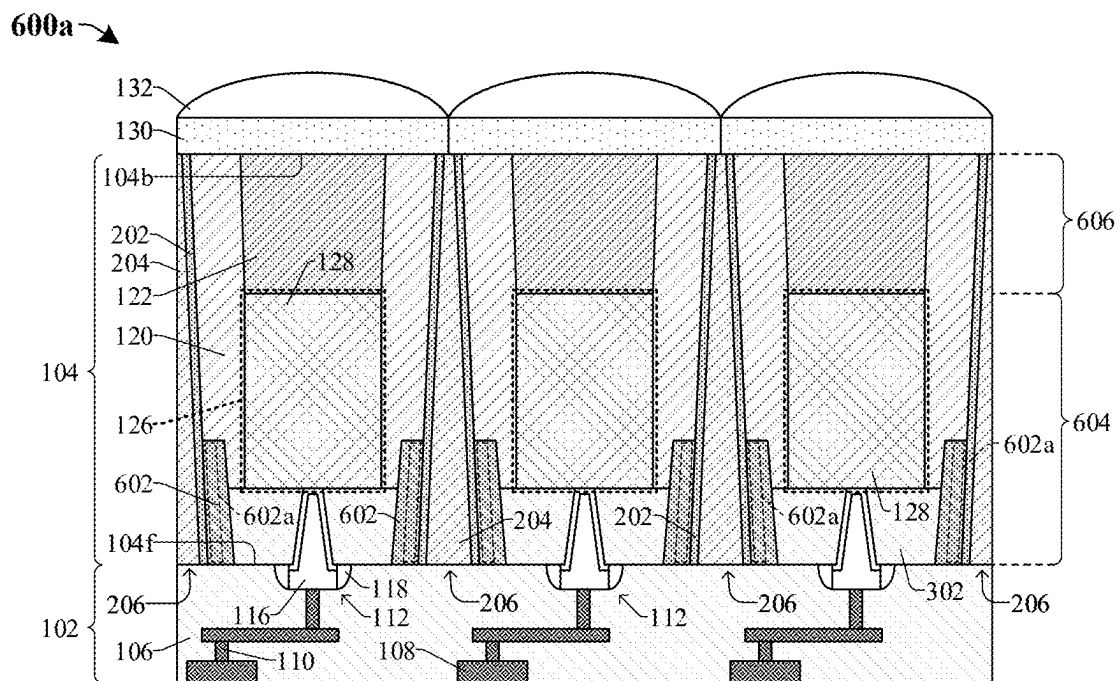


Fig. 6A

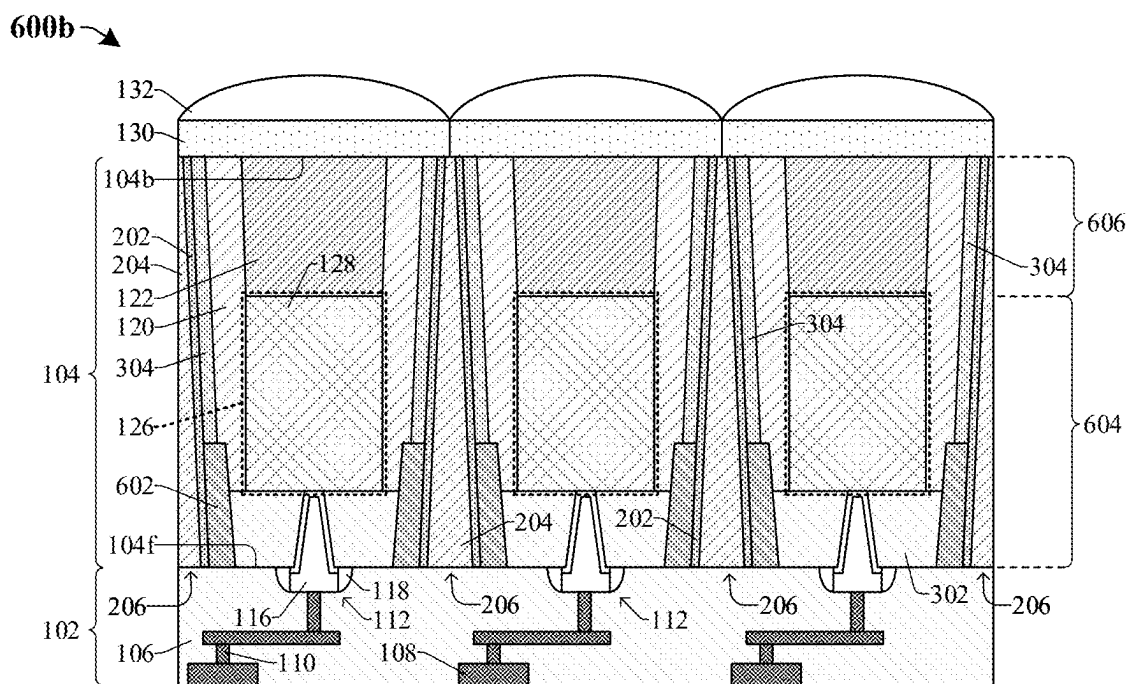


Fig. 6B

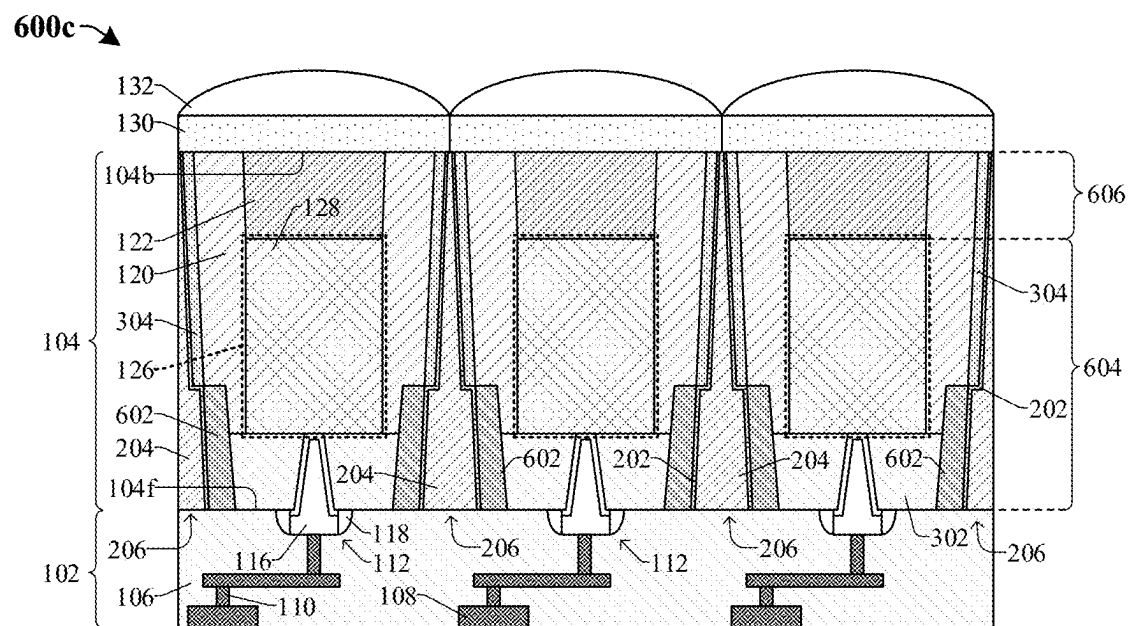


Fig. 6C

700

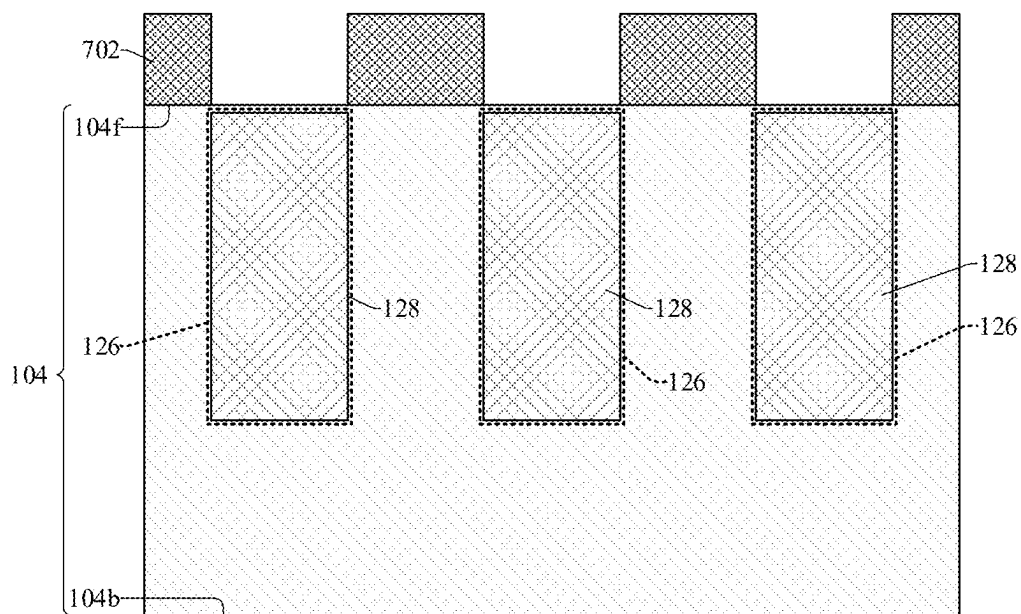


Fig. 7

800

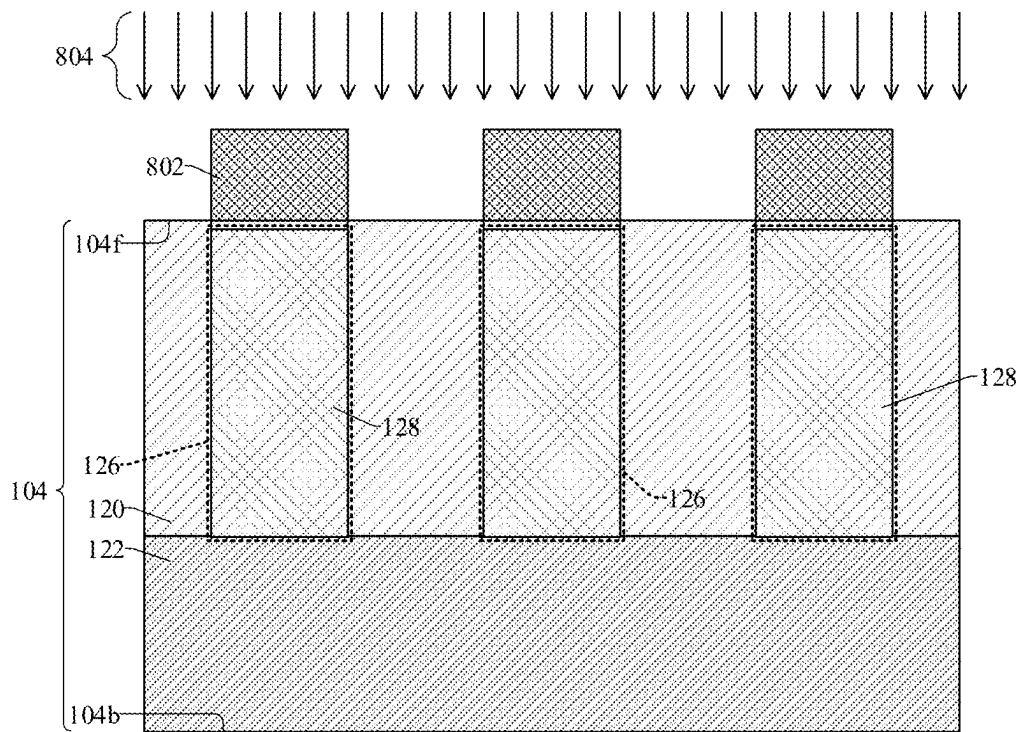


Fig. 8

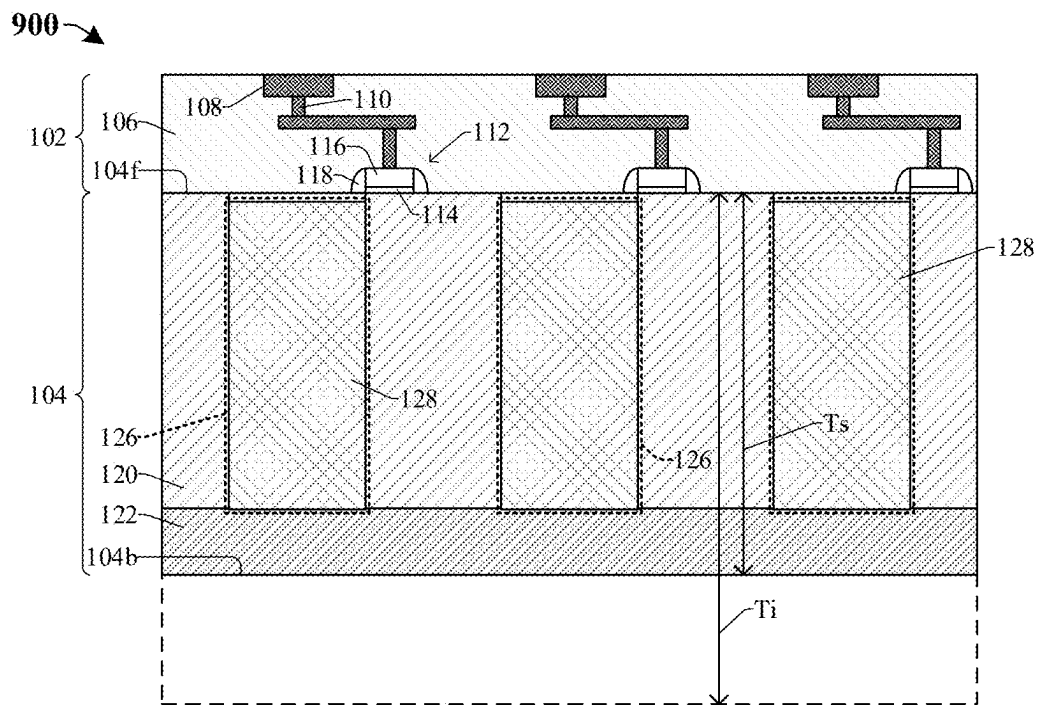


Fig. 9

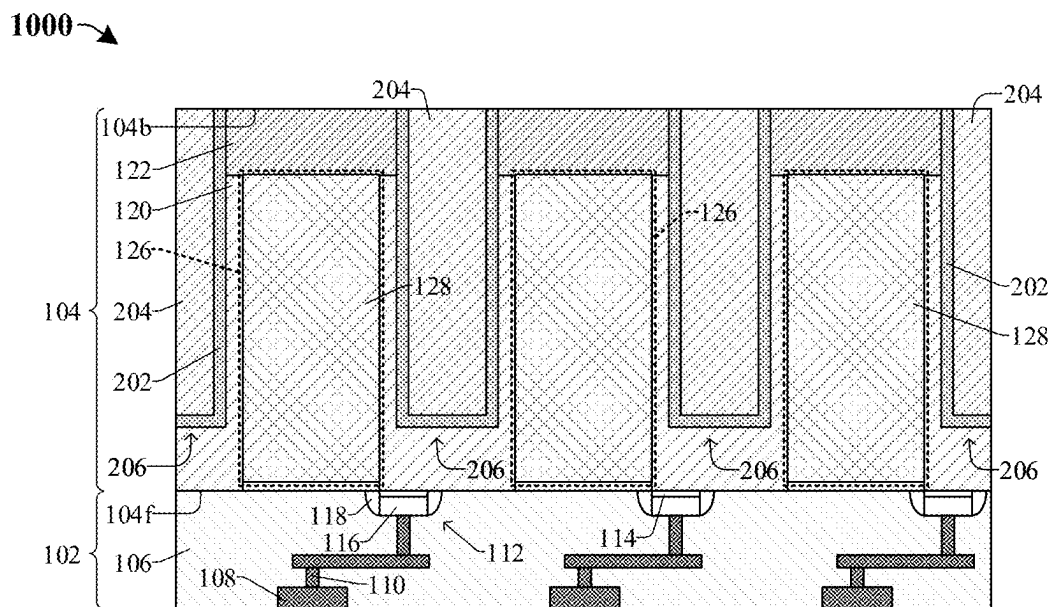


Fig. 10

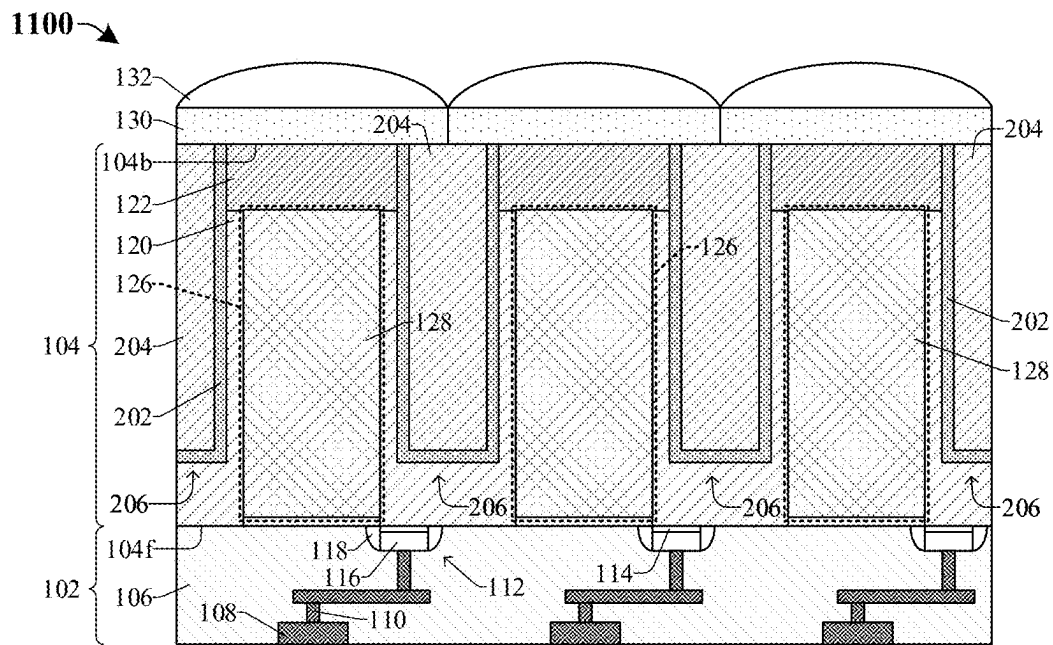


Fig. 11

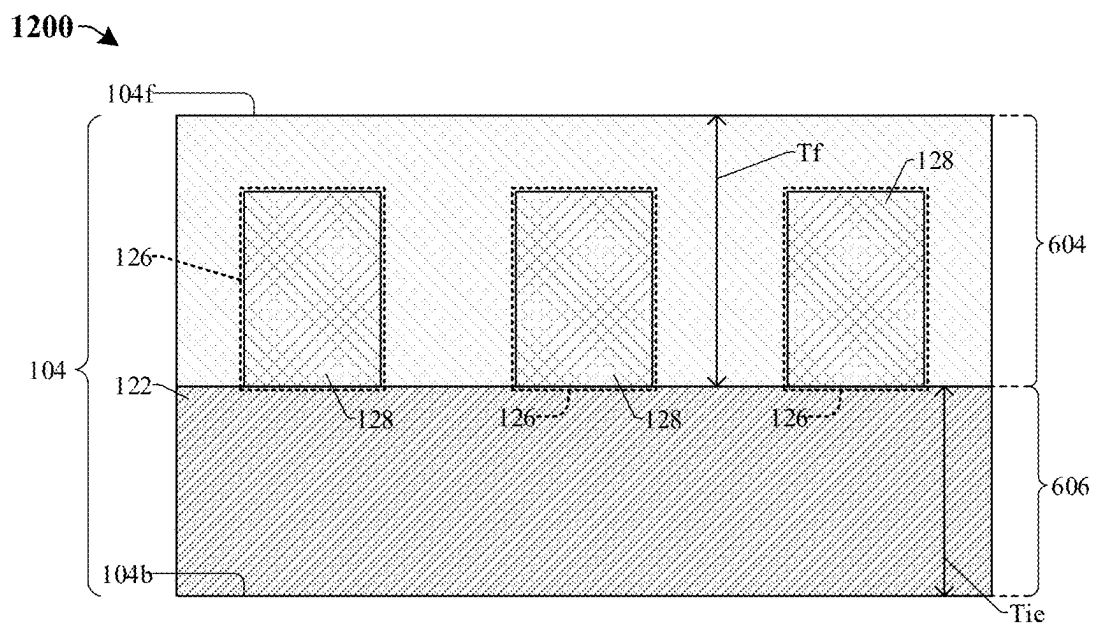


Fig. 12

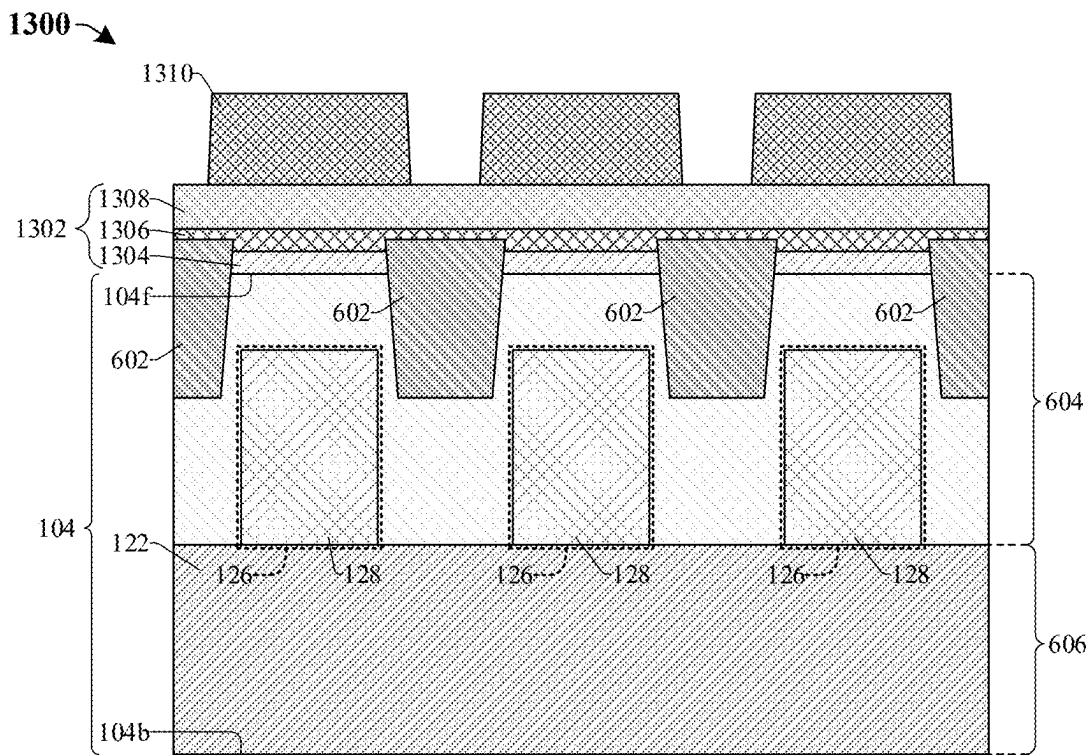


Fig. 13

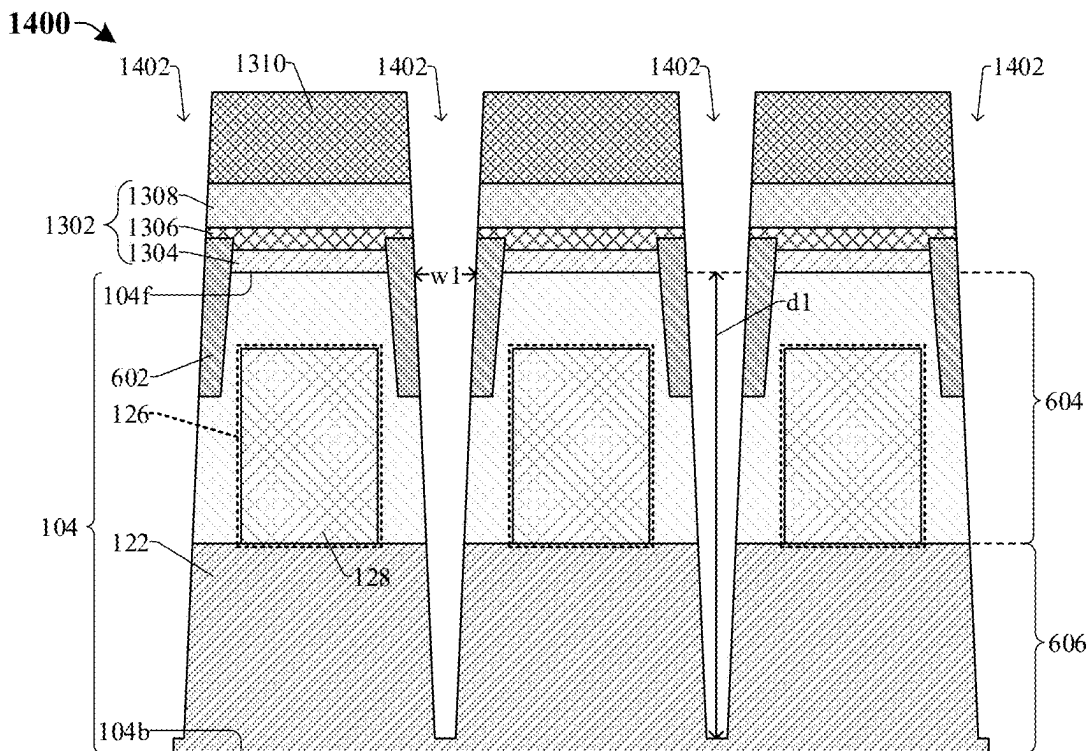
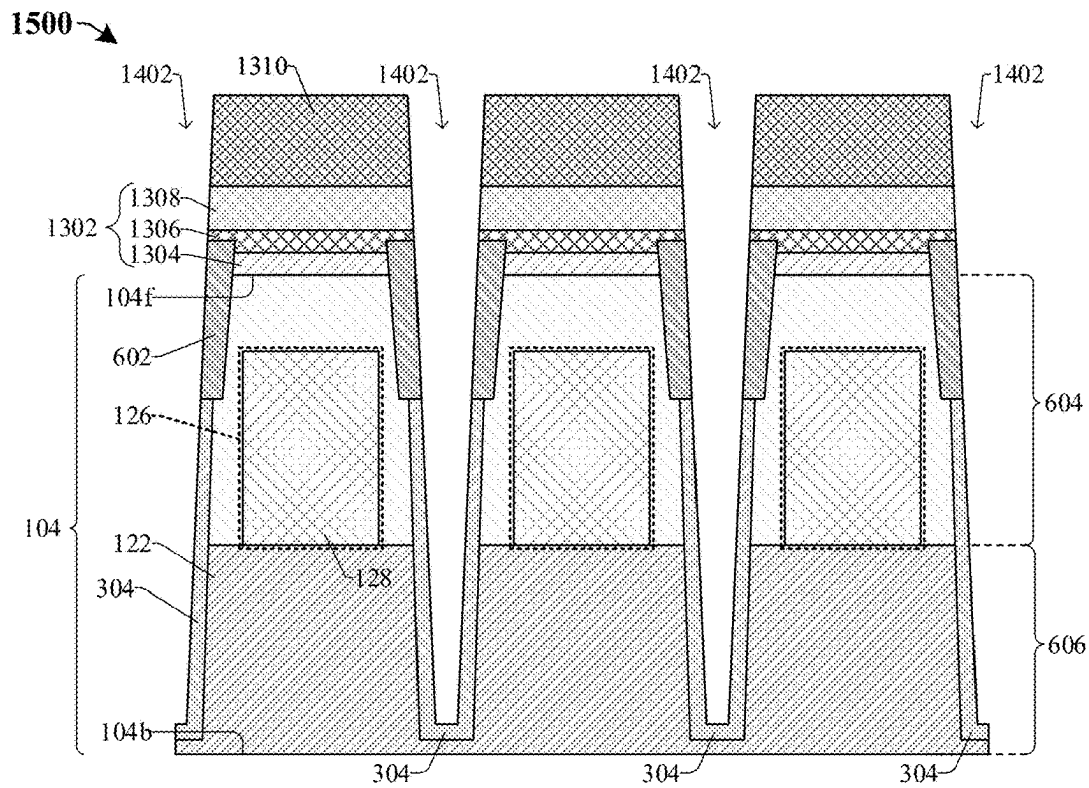
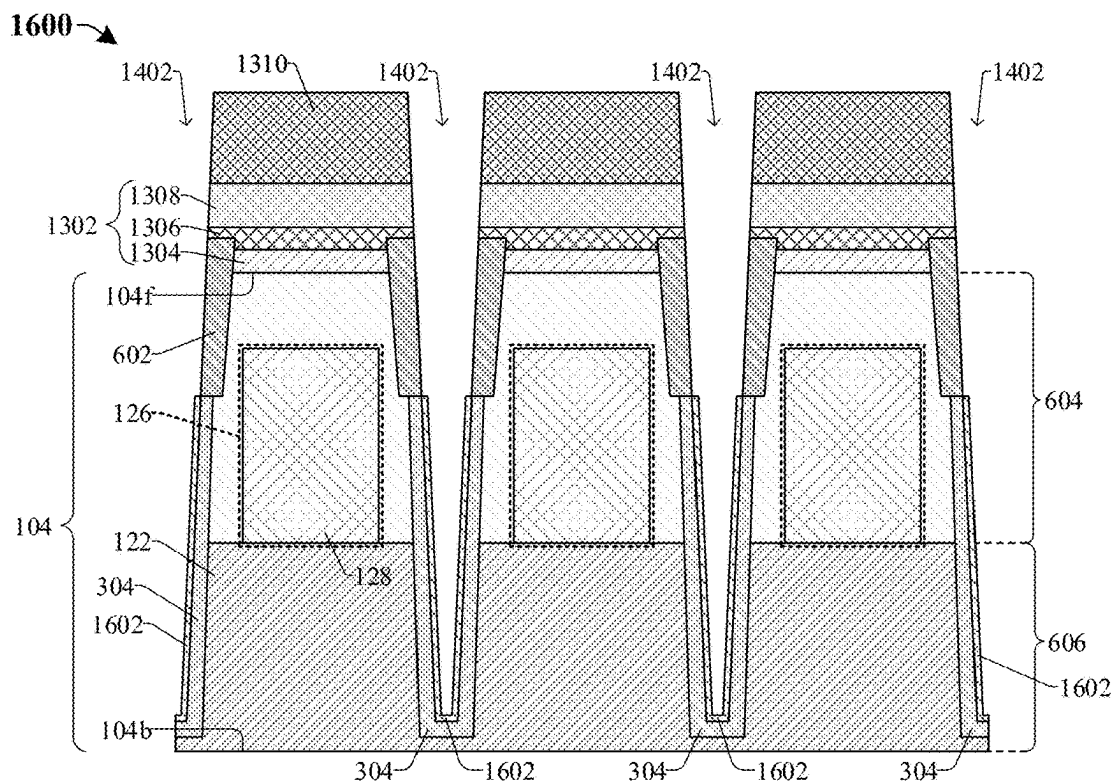


Fig. 14

**Fig. 15****Fig. 16**

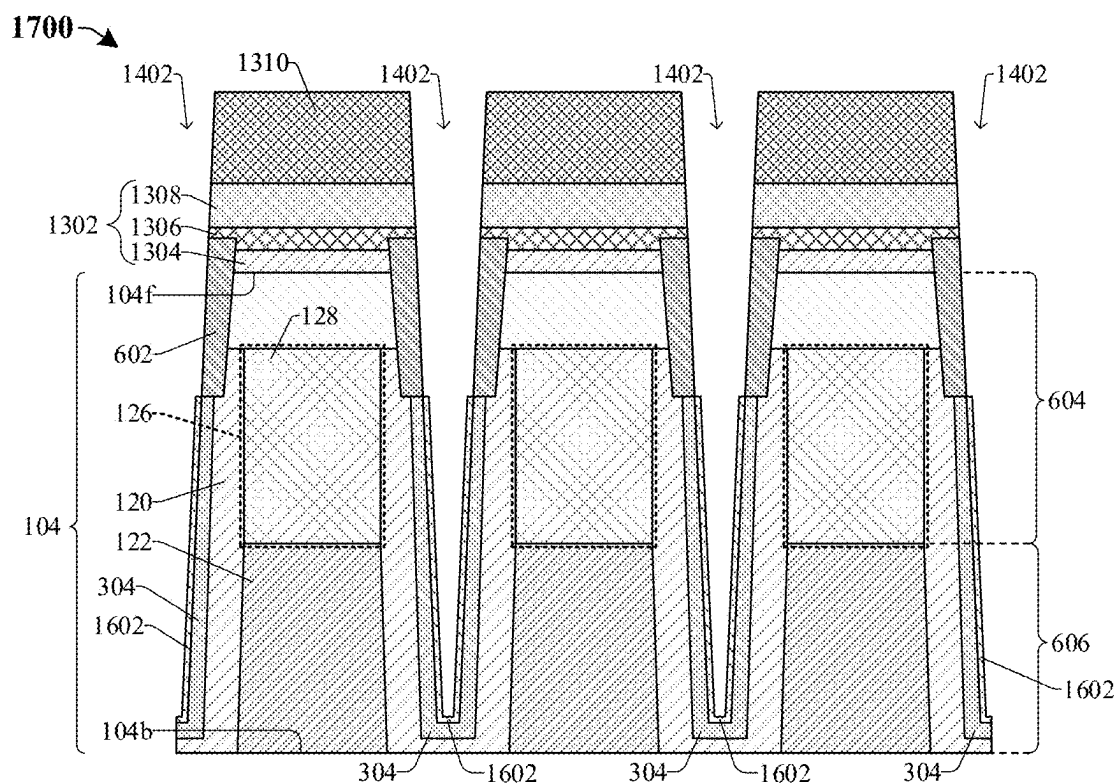


Fig. 17

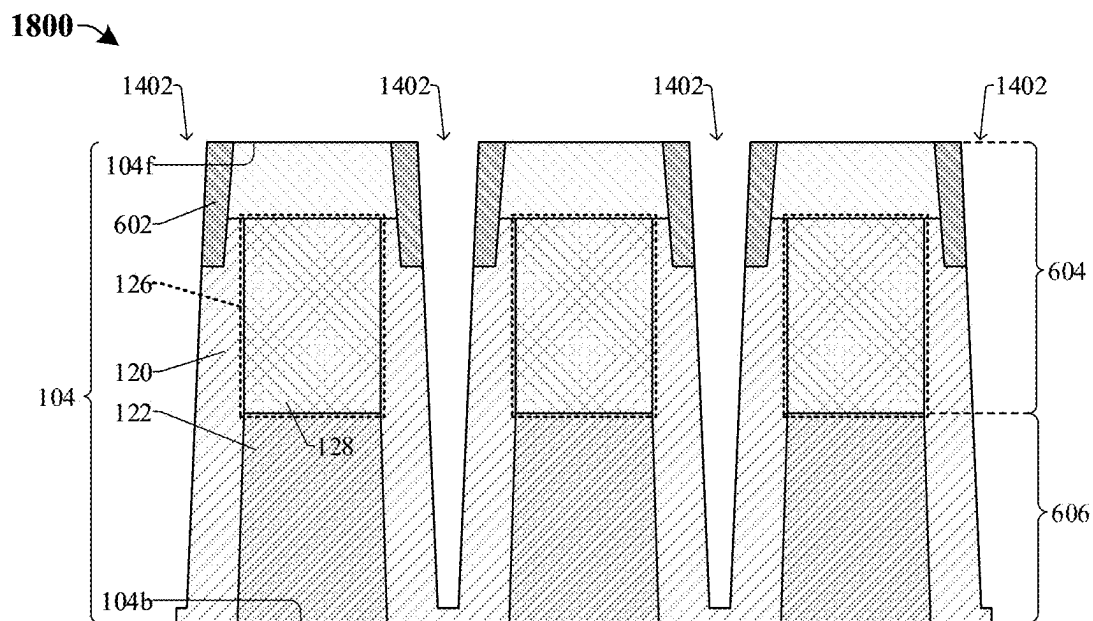


Fig. 18

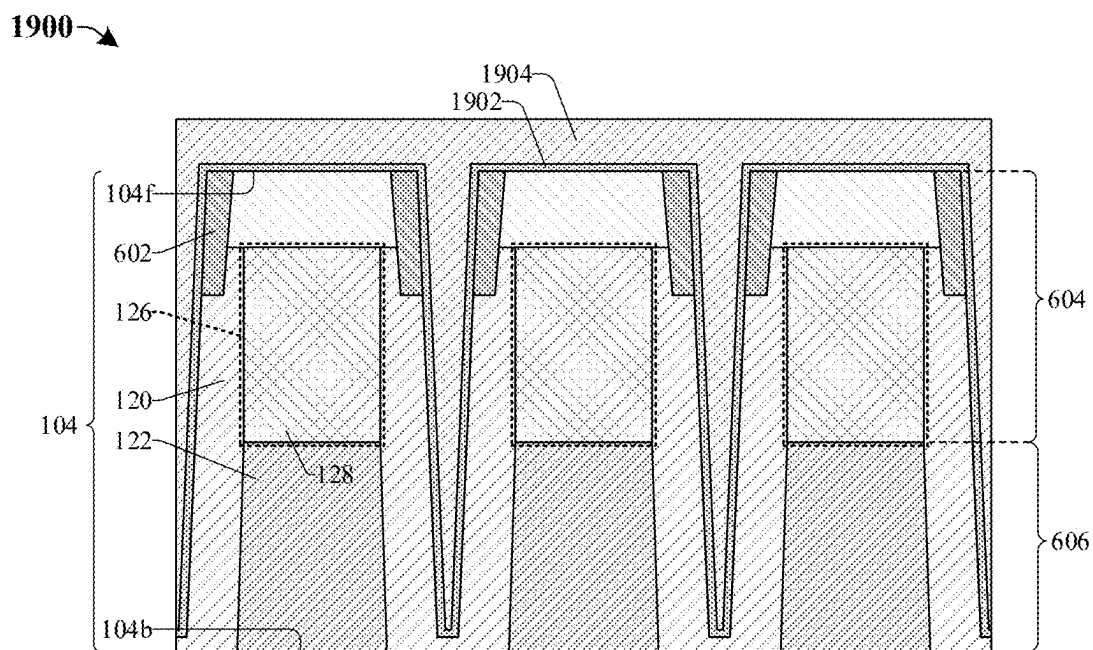


Fig. 19

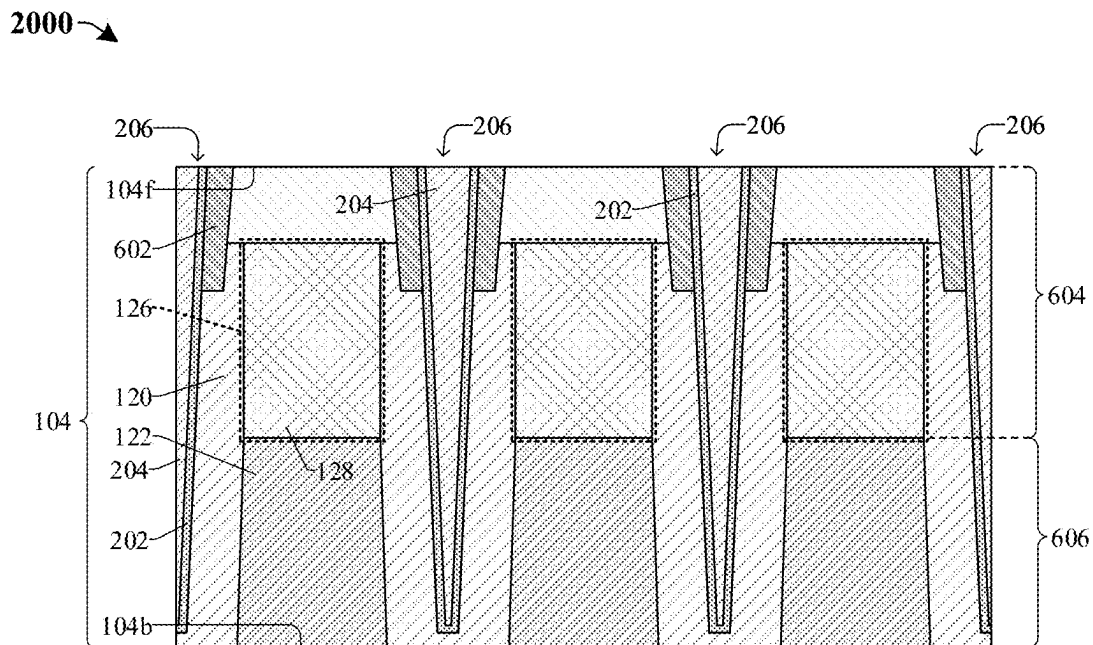


Fig. 20

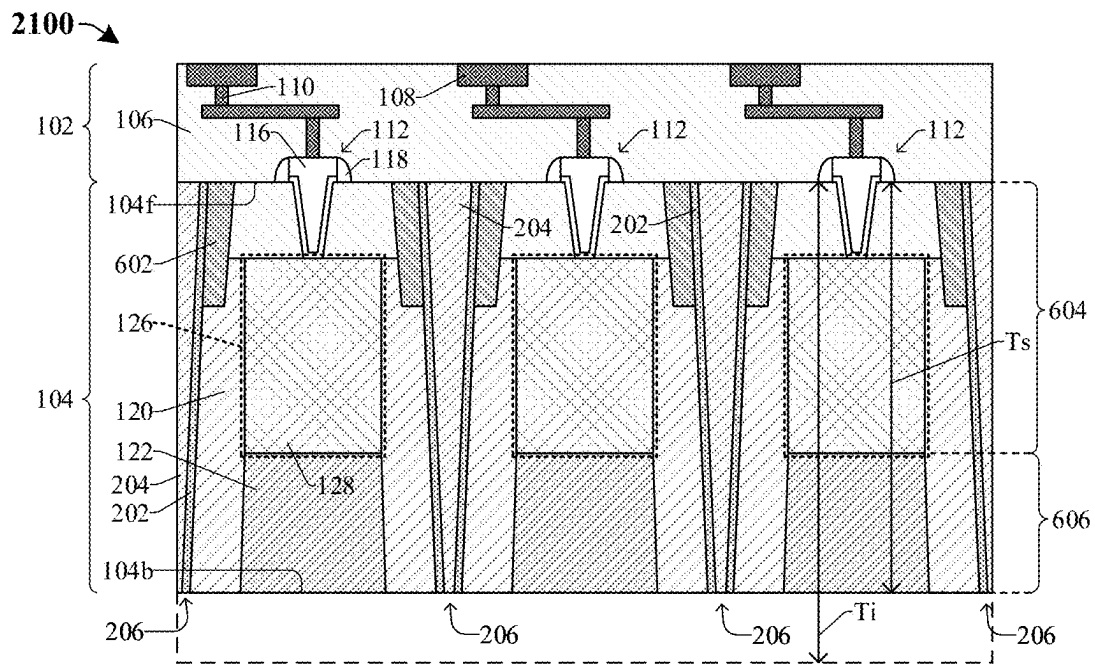


Fig. 21

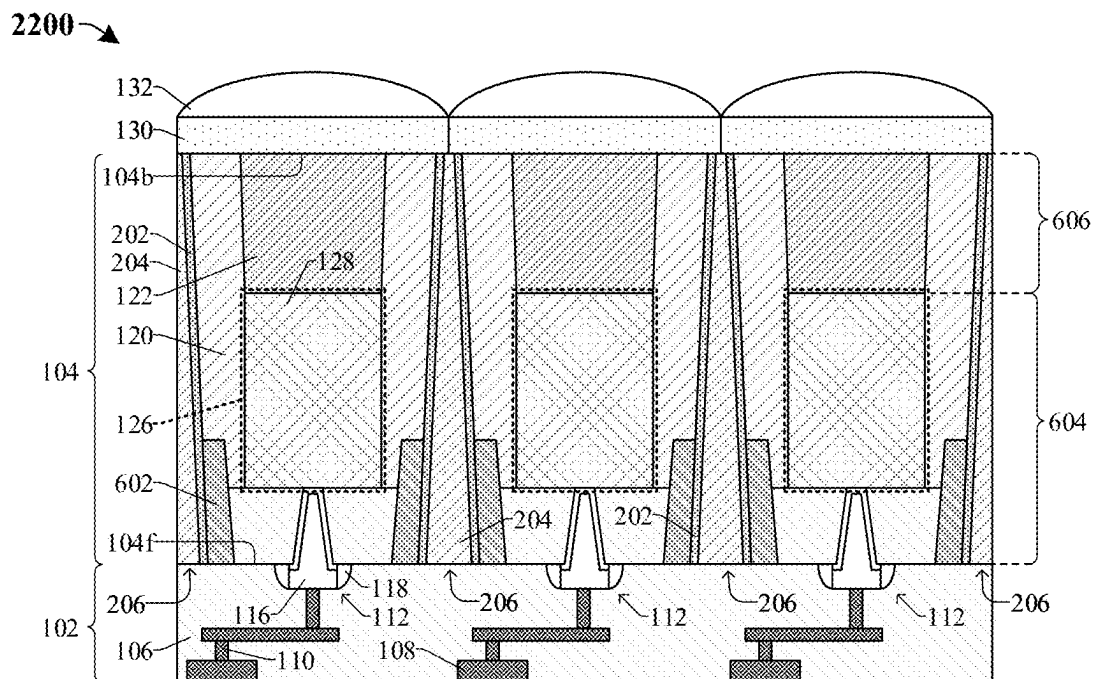


Fig. 22

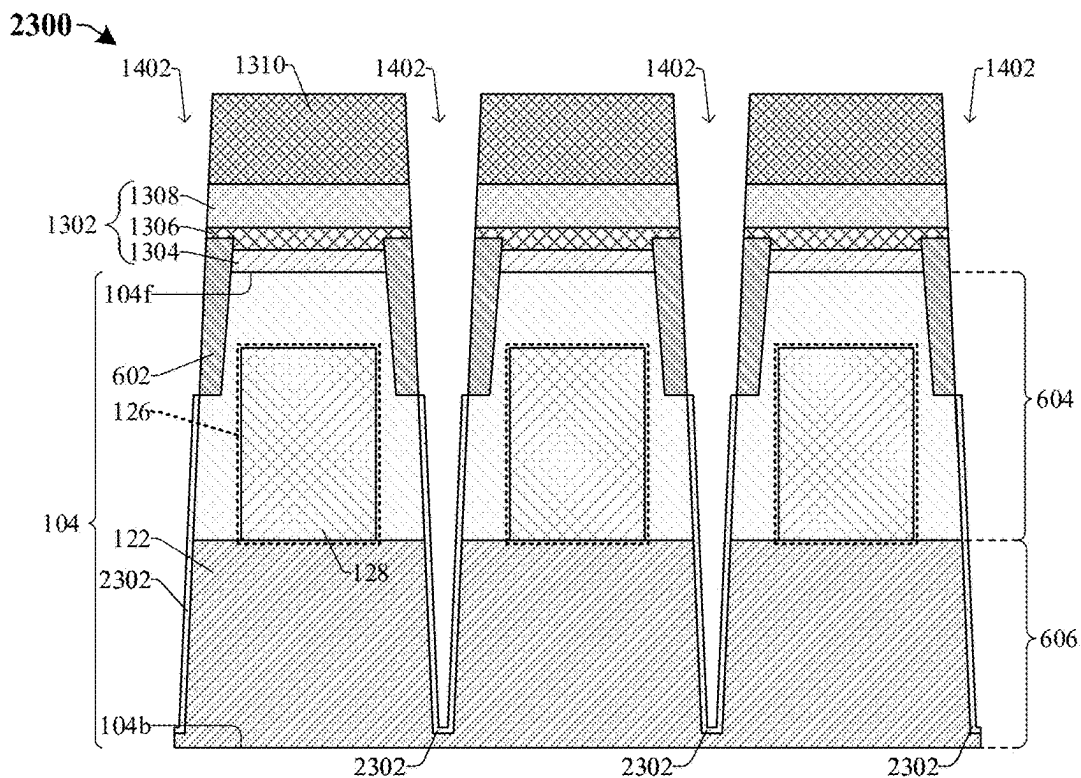


Fig. 23

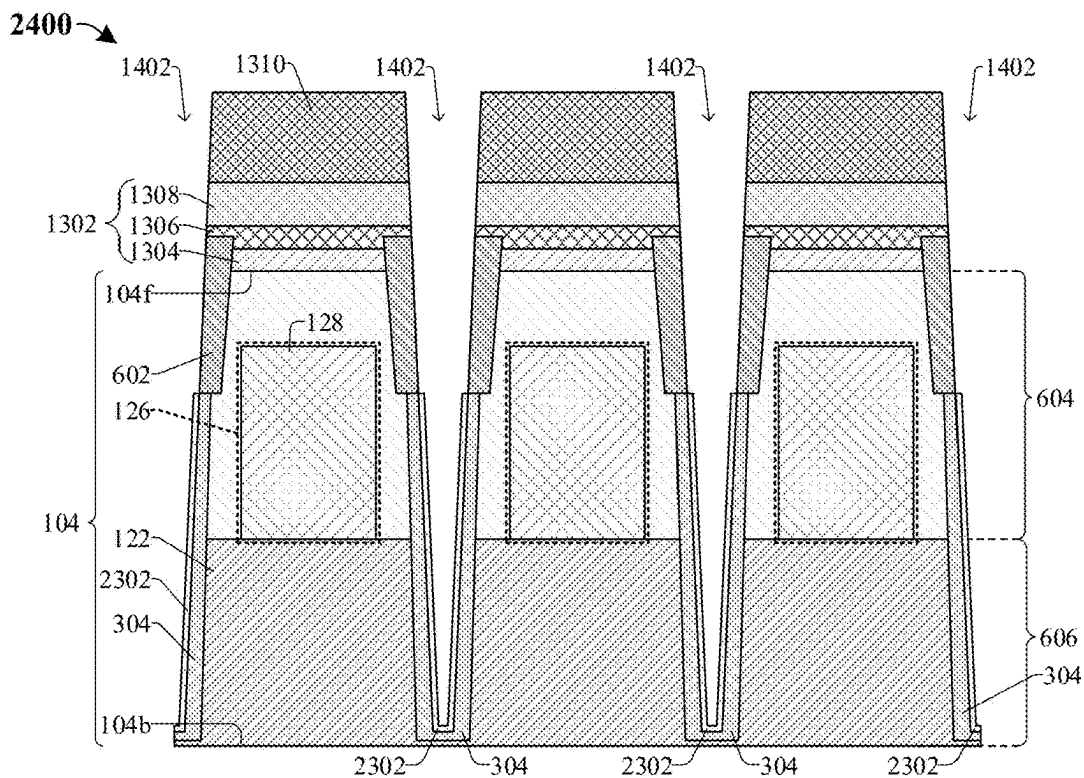


Fig. 24

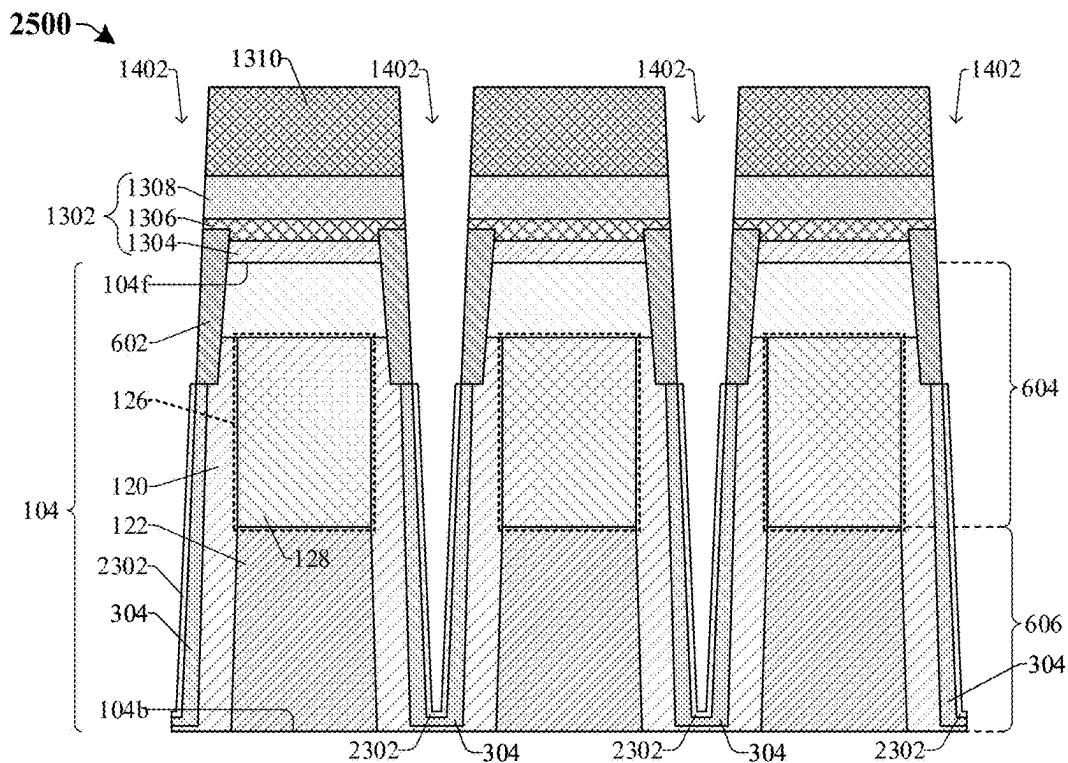


Fig. 25

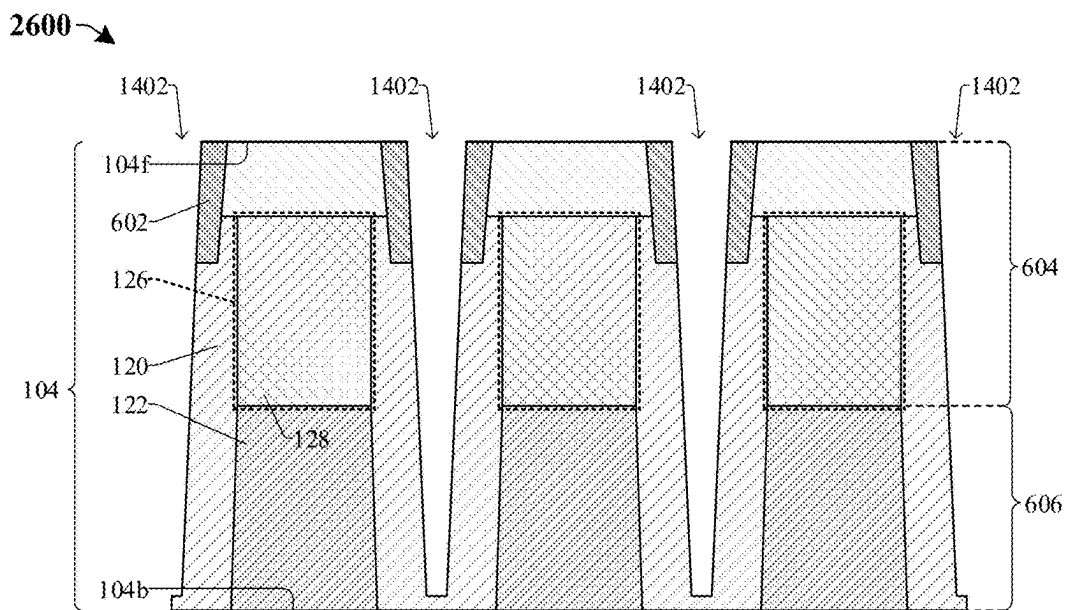


Fig. 26

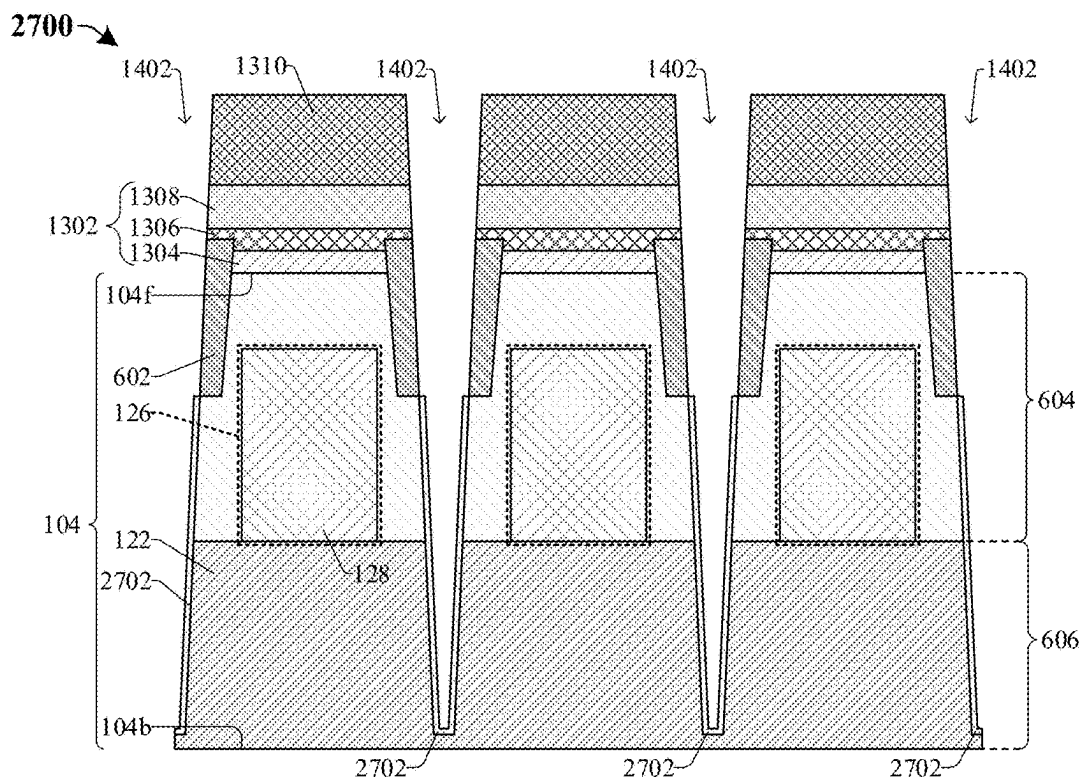


Fig. 27

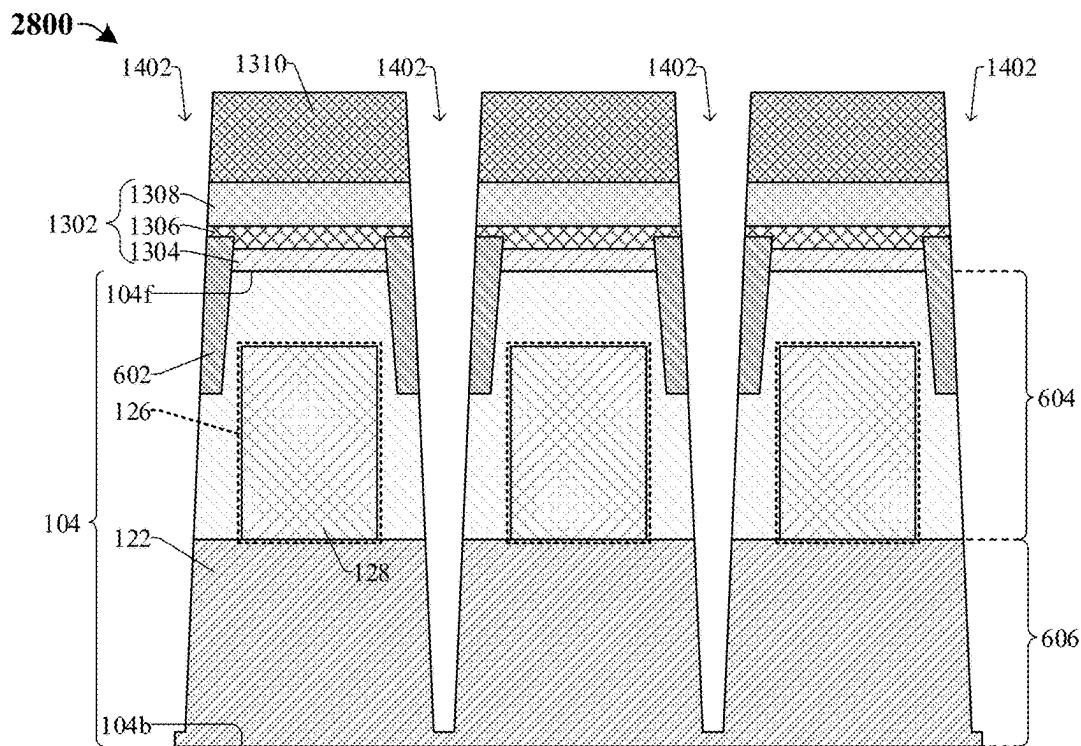


Fig. 28

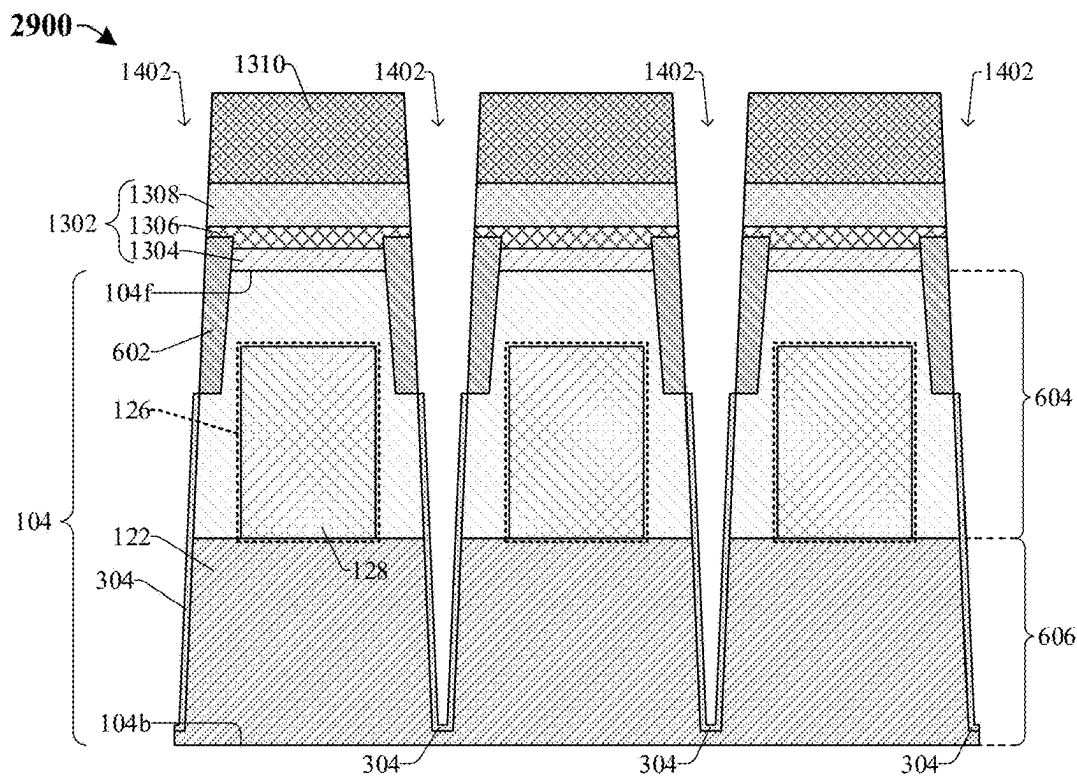


Fig. 29

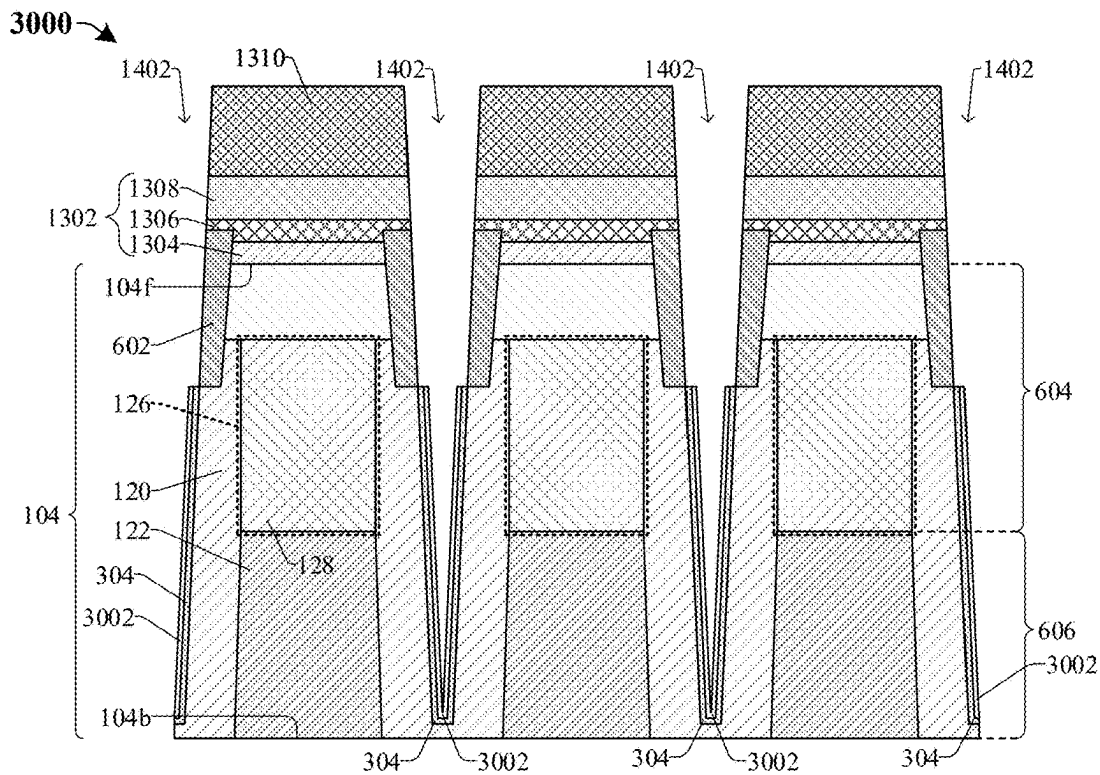


Fig. 30

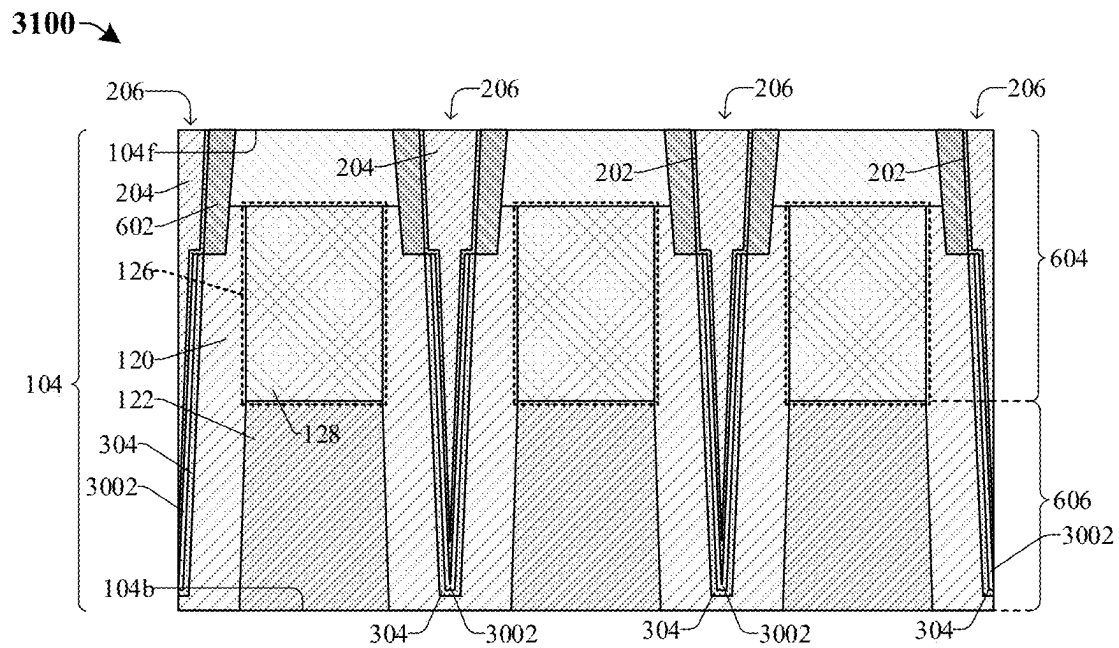
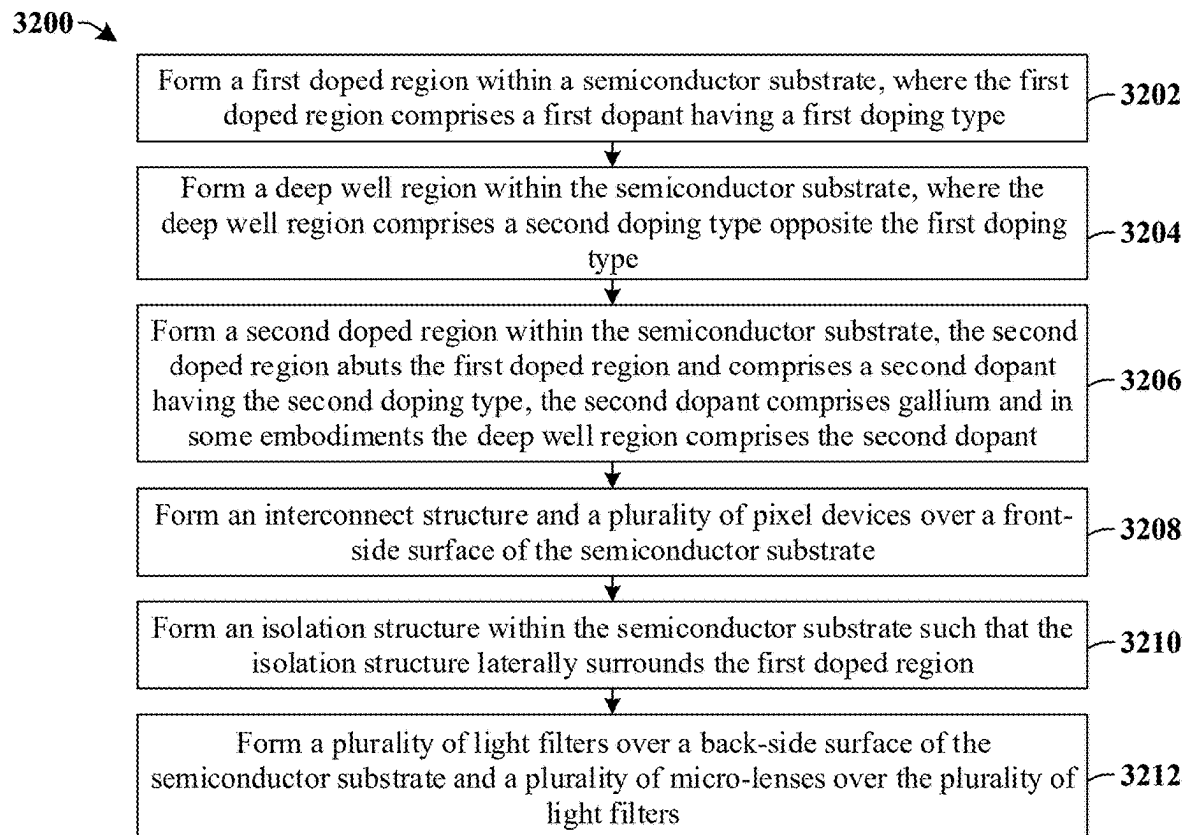


Fig. 31

**Fig. 32**

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FULL WELL CAPACITY FOR IMAGE SENSOR

REFERENCE TO RELATED APPLICATIONS

This Application is a Divisional of U.S. application Ser. No. 17/025,033, filed on Sep. 18, 2020, which claims the benefit of U.S. Provisional Application No. 62/982,559, filed on Feb. 27, 2020. The contents of the above-referenced Patent Applications are hereby incorporated by reference in their entirety.

BACKGROUND

Digital cameras and optical imaging devices employ image sensors. Image sensors convert optical images to digital data that may be represented as digital images. An image sensor typically includes an array of pixel sensors, which are unit devices for the conversion of an optical image into electrical signals. Pixel sensors often manifest as charge-coupled devices (CCDs) or complementary metal oxide semiconductor (CMOS) devices. However, CMOS pixel sensors have recently received more attention. Relative to CCD pixel sensors, CMOS pixel sensors provide lower power consumption, smaller size, and faster data processing. Further, CMOS pixel sensors provide a direct digital output of data, and generally have a lower manufacturing cost compared with CCD pixel sensors.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of some embodiments of an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors.

FIG. 2 illustrates a cross-sectional view of some embodiments of an image sensor comprising an isolation structure disposed between first doped regions, where a second doped region extends along sidewalls of the isolation structure.

FIGS. 3A-3C, 4A-4B, and 5A-5B illustrate cross-sectional views of some embodiments of image sensors according to some alternative embodiments of the image sensor of FIG. 2.

FIGS. 6A-6C illustrate cross-sectional views of some embodiments of image sensors comprising a plurality of first doped regions and an isolation structure laterally enclosing the first doped regions, where a second doped region is disposed between the isolation structure and the first doped regions.

FIGS. 7-11 illustrate cross-sectional views of some embodiments of a first method of forming an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors.

FIGS. 12-22 illustrate cross-sectional views of some embodiments of a second method of forming an image

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sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors.

FIGS. 23-26 illustrate cross-sectional views of some alternative embodiments of the second method of FIGS. 12-22.

FIGS. 27-31 illustrate cross-sectional views of some additional alternative embodiments of the second method of FIGS. 12-22.

FIG. 32 illustrates a flowchart according to some embodiments of a method of forming an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors.

DETAILED DESCRIPTION

The present disclosure provides many different embodiments, or examples, for implementing different features of this disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Many portable electronic devices (e.g., cameras, cellular telephones, etc.) include an image sensor for capturing images. One example of such an image sensor is a complementary metal-oxide-semiconductor (CMOS) image sensor including an array of active pixel sensors. Each of the active pixel sensors comprises a photodetector disposed in a semiconductor substrate. The photodetector comprises a p-n junction that exists between a first doped region, which has a first doping type (e.g., n-type doping), and a second doped region, which has a second doping type (e.g., p-type doping) opposite the first doping type.

In the case of the CMOS image sensor, out-diffusion of dopants within the second doped region may occur during front-end of line processing steps. For example, a process for forming the CMOS image sensor may include performing a first ion implantation process into a front-side surface of a semiconductor substrate to define the first doped region. Further, a second ion implantation process is performed into

the front-side surface of the semiconductor substrate to define the second doped region. The second ion implantation process may include implanting dopants (e.g., boron) that have a high likelihood to diffuse out of the second doped region during subsequent front-end of line processing steps (e.g., during formation of pixel devices (e.g., transistors) and/or an interconnect structure over the front-side surface). This may expand an area of the semiconductor substrate the second doped region occupies, thereby decreasing a size of the first doped region such that a volume of the photodetectors is reduced. Further, the out-diffusion of the dopants (e.g., boron) reduces a full well capacity (e.g., the amount of charge a photodetector can accumulate before saturation) of the photodetector because of the reduced size of the first doped region. Reducing the full well capacity of the photodetector may negatively affect the performance of the CMOS image sensor by, for example, reducing the high dynamic range of the active pixel sensors.

In various embodiments, the present disclosure is directed towards an image sensor that comprises a dopant (e.g., gallium), with a low likelihood to diffuse out, within the second doped region to improve full well capacity. In some embodiments, the image sensor has a photodetector having a first doped region disposed in a semiconductor substrate. The first doped region has a first doping type. A second doped region is disposed within the semiconductor substrate and abuts the first doped region. The second doped region comprises a second doping type opposite the first doping type. Further, the second doped region comprises the dopant (e.g., gallium) with relatively low diffusivity, thereby decreasing diffusion of the dopant from the second doped region during front-end of line processing steps and/or operation of the image sensor. This, in part, increases isolation between the first and second doped regions and increases the full well capacity of the photodetector, thereby improving performance of the image sensor.

FIG. 1 illustrates a cross-sectional view of some embodiments of an image sensor 100 comprising a plurality of photodetectors 126 having first doped regions 128 and a second doped region 120 laterally surrounding the first doped regions 128, where the second doped region 120 comprises a second dopant (e.g., gallium) configured to improve full well capacity of the photodetectors 126.

The image sensor 100 includes a semiconductor substrate 104 having a front-side surface 104f opposite a back-side surface 104b, and an interconnect structure 102 disposed along the front-side surface 104f. The semiconductor substrate 104 may, for example, be or comprise any type of semiconductor body (e.g., monocrystalline silicon, epitaxial silicon, CMOS bulk, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). The plurality of photodetectors 126 are disposed within the semiconductor substrate 104. In some embodiments, the plurality of photodetectors 126 are arranged in an array comprising a plurality of rows (e.g., along an x-axis) and columns (e.g., along a y-axis) of photodetectors. The photodetectors 126 are configured to convert incident radiation or incident light 134 (e.g., photons) and generate respective electrical signals corresponding to the incident light 134.

The interconnect structure 102 includes an interconnect dielectric structure 106, a plurality of conductive wires 108, and a plurality of conductive vias 110. Further, a plurality of pixel devices 112 are disposed within/on the front-side surface 104f of the semiconductor substrate 104. The pixel devices 112 each comprises a gate electrode 116, a gate dielectric layer 114, and a sidewall spacer structure 118. The

pixel devices 112 are configured to facilitate readout of the electrical signals produced by the photodetectors 126.

The photodetectors 126 each comprise a first doped region 128 (e.g., referred to as a photodetector collector region) comprising a first dopant having a first doping type (e.g., n-type). Further, a second doped region 120 (e.g., referred to as a photodetector well region) is disposed within the semiconductor substrate 104 and adjoins the first doped region 128. The second doped region 120 has a second doping type (e.g., p-type) that is opposite the first doping type. In some embodiments, the first doping type is n-type and the second doping type is p-type, or vice versa. In some embodiments, a depletion region forms (e.g., due to p-n junctions between the first doped region 128 and the second doped region 120) within each photodetector 126.

In addition, a deep well region 122 is disposed in the semiconductor substrate 104 and along the back-side surface 104b of the semiconductor substrate 104. The deep well region 122 extends into the semiconductor substrate 104 from the back-side surface 104b of the semiconductor substrate 104 to a point below the back-side surface 104b. In some embodiments, the deep well region 122 comprises the second doping type (e.g., p-type). A plurality of light filters 130 (e.g., color filters) are disposed over the back-side surface 104b of the semiconductor substrate 104. The plurality of light filters 130 are respectively configured to transmit specific wavelengths of incident light 134. A plurality of micro-lenses 132 are disposed over the plurality of light filters 130. The micro-lenses 132 are configured to focus the incident light 134 towards the photodetectors 126.

In some embodiments, the second doped region 120 and/or the deep well region 122 respectively comprise a second dopant having the second doping type (e.g., p-type) with a low likelihood to diffuse out. The second dopant may, for example, be or comprise gallium (Ga) that has a low likelihood to diffuse out of the second doped region 120 and/or the deep well region 122 during fabrication and/or operation of the image sensor 100. For example, during fabrication of the image sensor 100, the first doped region 128 may be formed within the semiconductor substrate 104. Subsequently, the second doped region 120 and/or the deep well region 122 may be formed by, for example, implanting the second dopant (e.g., Ga^{+3}) with a sufficiently large atomic size (e.g., about 62 picometers (pm)). In such embodiments, the second dopant facilitates the second doped region 120 and/or the deep well region 122 having sufficient hole concentration (e.g., at least 1×10^{18}) while mitigating diffusion of the second dopant across the lattice of the semiconductor substrate 104 to, for example, the first doped region 128. In some embodiments, if the second doped region 120 is formed with another dopant (e.g., boron (e.g., B^{3+})) having a relatively small atomic size (e.g., about 23 pm), then the another dopant (e.g., boron) may easily traverse the lattice structure of the semiconductor substrate 104 to the first doped region 128, thereby increasing a size of the second doped region 120 and decreasing a size of the first doped region 128. This, in part, reduces a full well capacity of the photodetectors 126, thereby adversely affecting performance of the image sensor 100 (e.g., reducing the high dynamic range of the image sensor 100). In yet further embodiments, if the second doped region 120 is formed with yet another dopant (e.g., indium (e.g., In^{3+})) having a relatively large atomic size (e.g., about 92 pm), then a solid solubility of the dopant (e.g., indium) is too low, thereby reducing a concentration of holes within the second doped region 120 and adversely affecting performance of the image sensor 100. Thus, by virtue of the second doped region 120

and/or the deep well region 122 respectively comprising the second dopant (e.g., gallium) with a low likelihood to diffuse out the full well capacity of the photodetectors 126 is increased, thereby increasing a performance of the image sensor 100.

In yet further embodiments, a diffusivity of the second dopant (e.g., gallium) at relatively high temperatures (e.g., about 700 degrees Celsius or greater) is relatively low (e.g., less than about 5.4×10^{-15} cm²/s). This, in part, mitigates diffusion of the second dopant during front-end of line formation processes and/or during operation of the image sensor 100. In another embodiment, if the second doped region 120 comprises another dopant (e.g., boron) with a relatively high diffusivity (e.g., about 1.7×10^{-14} cm²/s or greater) at the relatively high temperatures, then a greater concentration of the another dopant (e.g., boron) diffuses into the first doped regions 128 during front-end of line formation processes and/or operation of the image sensor 100. This decreases a hole concentration in the second doped region 120, decreases isolation between photodetectors 126, and decreases the full well capacity of the photodetectors 126. Further, in an effort to increase a number of semiconductor devices disposed within/or on the semiconductor substrate 104, a size of the photodetectors 126 may be decreased. The relatively low diffusivity of the second dopant (e.g., gallium) promotes electrical isolation between the photodetectors 126 as the feature size of the photodetectors 126 is decreased, thereby increasing performance of the image sensor 100.

In addition, a depletion region forms along the perimeter of each first doped region 128 (e.g., due to p-n junctions between the first doped region 128 and the second doped region 120 and/or the deep well region 122). In some embodiments, when the incident light 134 (containing photons of sufficient energy) strikes a corresponding photodetector 126, an electron-hole pair is created. If the absorption occurs in the junction's depletion region (e.g., one or more diffusion length(s) away from it) the carriers of the electron-hole pair are swept from the junction by the built-in electric field of the depletion region. Thus, holes move towards an anode region of the corresponding photodetector 126 and electrons move toward a cathode region of the corresponding photodetector 126, thereby producing a photocurrent. Due to the relatively low diffusivity of the second dopant (e.g., gallium), a strength of the built-in electric field of the depletion region is increased, thereby ensuring the electrons move toward the cathode region of the photodetector 126. This, in part, increases isolation between adjacent photodetectors 126 and prevents the recombination of charge carriers (e.g., electrons) at sidewalls of an isolation structure (e.g., sidewalls of the isolation structure 206 of FIG. 2).

FIG. 2 illustrates a cross-sectional view of some embodiments of an image sensor 200 corresponding to some alternative embodiments of the image sensor 100 of FIG. 1.

The image sensor 200 includes an interconnect structure 102 disposed along a front-side surface 104f of a semiconductor substrate 104. In some embodiments, the semiconductor substrate 104 may, for example, be or comprise a bulk substrate (e.g., a bulk silicon substrate), a silicon on insulator substrate, crystalline silicon, monocrystalline silicon, epitaxial silicon, doped epitaxial silicon, an oxygen free silicon substrate, an oxygen rich silicon substrate, P-doped silicon, or another suitable semiconductor material. The interconnect structure 102 includes an interconnect dielectric structure 106, a plurality of conductive wires 108, and a plurality of conductive vias 110. In some embodiments, the interconnect dielectric structure 106 comprises one or more

inter-metal dielectric (IMD) layers. The interconnect structure 102 is configured to electrically couple semiconductor devices disposed within and/or on the semiconductor substrate 104 to one another. In further embodiments, the conductive wires 108 and/or the conductive vias 110 may, for example, be or comprise aluminum, copper, tungsten, titanium, tantalum, titanium nitride, tantalum nitride, ruthenium, another conductive material, or any combination of the foregoing. In yet further embodiments, the IMD layers may, for example, be or comprise an oxide such as silicon dioxide, a low-k dielectric material, an extreme low-k dielectric material, another dielectric material, or any combination of the foregoing. As used herein, a low-k dielectric material is a dielectric material with a dielectric constant less than 3.9.

Further, a plurality of pixel devices 112 are disposed along the front-side surface 104f of the semiconductor substrate 104. The pixel devices 112 can, for example, be configured as transistors and comprise a gate electrode 116, a gate dielectric layer 114, and a sidewall spacer structure 118. The gate dielectric layer 114 is disposed between the gate electrode 116 and the semiconductor substrate 104. Further, the sidewall spacer structure 118 is disposed along sidewalls of the gate electrode 116 and sidewalls of the gate dielectric layer 114. The plurality of pixel devices 112 may, for example, be or comprise transfer transistor(s), source-follower transistor(s), row-select transistor(s), reset transistor(s), another suitable pixel device, or any combination of the foregoing. The plurality of pixel devices 112 are electrically coupled to the conductive wires and vias 108, 110. The gate electrode 116 may, for example, be or comprise polysilicon, doped polysilicon, a metal material such as aluminum, copper, titanium, tantalum, tungsten, titanium nitride, tantalum nitride, another suitable material, or any combination of the foregoing. The gate dielectric layer may, for example, silicon dioxide, a high-k dielectric material, a combination of the foregoing, or the like. As used herein, a high-k dielectric material is a dielectric material with a dielectric constant greater than 3.9. Further, the sidewall spacer structure 118 may, for example, be or comprise silicon nitride, silicon carbide, another dielectric material, or any combination of the foregoing.

A plurality of photodetectors 126 are disposed within the semiconductor substrate 104. The photodetectors 126 each comprise a first doped region 128 having a first doping type (e.g., n-type) with a doping concentration of the ranging between about 10^{14} to 10^{18} atoms/cm³, or another suitable value. Further, a second doped region 120 is disposed within the semiconductor substrate 104 and laterally surrounds the first doped region 128. The second doped region 120 has a second doping type (e.g., p-type) that is opposite the first doping type. In various embodiments, a doping concentration of the first doped region 128 is less than a doping concentration of the second doped region 120. Further, a deep well region 122 is disposed within the semiconductor substrate 104 along a back-side surface 104b of the semiconductor substrate 104. The deep well region 122 extends from the back-side surface 104b of the semiconductor substrate 104 to a top surface of the first doped region 128. The deep well region 122 comprises the second doping type (e.g., p-type). In some embodiments, the second doped region 120 and the deep well region 122 comprise a second dopant having the second doping type (e.g., p-type) with a low likelihood to diffuse out. The second dopant may, for example, be or comprise gallium (Ga) that has a low likelihood to diffuse out of the second doped region 120 and/or the deep well region 122 during fabrication and/or

operation of the image sensor **100**. This, in part, increases a full well capacity of the photodetectors **126**, thereby increasing a performance of the image sensor **200**. It will be appreciated that the second dopant comprising another element is also within the scope of the disclosure.

In some embodiments, the second doped region **120** and the deep well region **122** have a first doping concentration of the second dopant (e.g., gallium) ranging between about 10^{14} to 2×10^{18} atoms/cm³, or another suitable value. In further embodiments, if the first doping concentration of the second dopant is relatively low (e.g., less than about 10^{14} atoms/cm³), then a concentration of holes may be relatively low, thereby decreasing electrical isolation between the plurality of photodetectors **126**. In yet further embodiments, if the first doping concentration of the second dopant is relatively high (e.g., greater than about 2×10^{18} atoms/cm³), then a maximum solid solubility of the second dopant within the semiconductor substrate **104** may be surpassed. In such embodiments, solid precipitates may form in the semiconductor substrate **104** that cause defects in the semiconductor substrate **104** and decrease a performance of the photodetectors **126**. In some embodiments, a doping concentration of the second doped region **120** and/or the deep well region **122** is greater than a doping concentration of the first doped region **128**.

In addition, an isolation structure **206** extends from the back-side surface **104b** of the semiconductor substrate **104** to a point below the back-side surface **104b**. The isolation structure **206** can, for example, be configured as a deep trench isolation (DTI) structure, a back-side deep trench isolation (BDTI) structure, a front-side deep trench isolation (FDTI) structure, or another suitable isolation structure. The isolation structure **206** is disposed between adjacent photodetectors **126** and is configured to electrically isolate the photodetectors **126** from one another. Further, the isolation structure **206** comprises a passivation layer **202** a trench fill layer **204**. The passivation layer **202** is disposed between the trench fill layer **204** and the semiconductor substrate **104**. The passivation layer **202** may, for example, be or comprise aluminum oxide (Al₂O₃), hafnium oxide (HfO₂), titanium oxide (Ta₂O₅), another high-k dielectric material, any combination of the foregoing, or the like. The trench fill layer **204** may, for example, be or comprise silicon dioxide, polysilicon, amorphous silicon, doped polysilicon, another dielectric material, or any combination of the foregoing. In yet further embodiments, a refractive index of the trench fill layer **204** is less than a refractive index of the semiconductor substrate **104**.

A plurality of light filters **130** (e.g., color filters) are disposed over the back-side surface **104b** of the semiconductor substrate **104**. The light filters **130** are respectively configured to transmit specific wavelengths of incident light **134**. For example, a first light filter (e.g., a red light filter) may transmit light having wavelengths within a first range, while a second light filter (e.g., a green light filter) may transmit light having wavelengths within a second range different than the first range. In some embodiments, the plurality of light filters **130** may be arranged within a grid structure overlying the photodetectors **126**. A plurality of micro-lenses **132** are arranged over the plurality of light filters **130**. In some embodiments, the plurality of micro-lenses **132** have a substantially flat bottom surface abutting the plurality of light filters **130** and a curved upper surface. The curved upper surface is configured to focus the incident light **134** towards the underlying photodetectors **126**.

FIG. 3A illustrates a cross-sectional view of some embodiments of an image sensor **300a** corresponding to

some alternative embodiments of the image sensor **200** of FIG. 2, in which the pixel devices **112** are configured as vertical transistors. In such embodiments, the pixel devices **112** may be configured as vertical transfer transistors, where the gate electrode **116** extends continuously from the front-side surface **104f** of the semiconductor substrate **104** to a point above the front-side surface **104f**. In some embodiments, the point is disposed above a bottom surface of the first doped region **128** such that the gate electrode **116** extends into the first doped region **128**. Further, the gate dielectric layer **114** is disposed between the gate electrode **116** and the first doped region **128**. A third doped region **302** of the semiconductor substrate **104** is disposed between the first doped region **128** and the front-side surface **104f** of the semiconductor substrate **104**. In yet further embodiments, the third doped region **302** comprises the second doping type (e.g., p-type). In yet further embodiments, the third doped region **302** comprises another dopant (e.g., boron) with a doping concentration within a range of about 10^{14} to 5×10^{15} atoms/cm³, or another suitable value. In various embodiments, a doping concentration of the third doped region **302** is less than a doping concentration of the second doped region **120**. Further, the second doped region **120** continuously extends along opposing sidewalls of the passivation layer **202** from the back-side surface **104b** of the semiconductor substrate **104** to a bottom surface of the passivation layer **202**.

During operation of the image sensor **300a**, the pixel devices **112** may, for example, be configured to transfer charge collected in a corresponding photodetector **126** to a floating node (not shown) disposed within the semiconductor substrate **104**. If the charge level is sufficiently high within the floating node, a source-follower transistor (not shown) is activated and charge is selectively output according to operation of a row-select transistor (not shown) used for addressing. A reset transistor (not shown) can be used to reset the photodetectors **126** between exposure periods.

FIG. 3B illustrates a cross-sectional view of some embodiments of an image sensor **300b** corresponding to some alternative embodiments of the image sensor **300a** of FIG. 3A, in which the isolation structure **206** extends continuously from the back-side surface **104b** of the semiconductor substrate **104** to the front-side surface **104f** of the semiconductor substrate **104**.

FIG. 3C illustrates a cross-sectional view of some embodiments of an image sensor **300c** corresponding to some alternative embodiments of the image sensor **300a** of FIG. 3A, in which a doped liner **304** is disposed between the passivation layer **202** and the semiconductor substrate **104**. In some embodiments, the doped liner **304** comprises the second dopant (e.g., gallium) with a doping concentration within a range of about between about 10^{14} to 2×10^{18} atoms/cm³, or another suitable value. In yet further embodiments, the doped liner **304** may facilitate formation of the second doped region **120** (e.g., see FIGS. 15-18). In some embodiments, the doped liner **304** may be a doped region of the semiconductor substrate **104** formed by a plasma doping process. In an alternative embodiment, the doped liner **304** may be or comprise an epitaxial silicon layer formed by an epitaxial process.

FIG. 4A illustrates a cross-sectional view of some embodiments of an image sensor **400a** corresponding to some alternative embodiments of the image sensor **200** of FIG. 2, in which the passivation layer **202** and the trench fill layer **204** continuously extend along the back-side surface **104b** of the semiconductor substrate **104**.

FIG. 4B illustrates a cross-sectional view of some embodiments of an image sensor **400b** corresponding to some alternative embodiments of the image sensor **400a** of FIG. 4A, in which the doped liner **304** is disposed between the semiconductor substrate **104** and the passivation layer **202**. In such embodiments, the doped liner **304** continuously extends along the back-side surface **104b** of the semiconductor substrate **104**.

FIG. 5A illustrates a cross-sectional view of some embodiments of an image sensor **500a** corresponding to some alternative embodiments of the image sensor **200** of FIG. 2, in which a bottom surface of the isolation structure **206** is curved. Further, the passivation layer **202** continuously extends along the back-side surface **104b** of the semiconductor substrate **104**.

FIG. 5B illustrates a cross-sectional view of some embodiments of an image sensor **500b** corresponding to some alternative embodiments of the image sensor **500b** of FIG. 5B, in which the doped liner **304** is disposed between the semiconductor substrate **104** and the passivation layer **202**.

FIG. 6A illustrates a cross-sectional view of some embodiments of an image sensor **600a** comprising a plurality of first doped regions **128** and an isolation structure **206** laterally enclosing the first doped regions **128**, where a second doped region **120** comprising a second dopant (e.g., gallium) is disposed between the isolation structure **206** and the first doped regions **128**.

In some embodiments, the image sensor **600a** comprises a shallow trench isolation (STI) structure **602** disposed within the semiconductor substrate **104**. In some embodiments, the semiconductor substrate **104** may, for example, be or comprise any type of semiconductor body (e.g., silicon/germanium/CMOS bulk, SiGe, SOI, etc.) such as a semiconductor wafer or one or more die on a wafer, as well as any other type of semiconductor and/or epitaxial layers. For example, the semiconductor substrate **104** may be or comprise a first epitaxial layer **604** underlying a second epitaxial layer **606**. In some embodiments, the first epitaxial layer **604** may be or comprise a p-type epitaxial silicon layer having a thickness of about 5.4 micrometers (um), about 6 um, within a range of about 5.4 to 6 um, or another suitable value. In yet further embodiments, the first epitaxial layer **604** may be or comprise an n-type epitaxial silicon layer having a thickness of about 11 um, within a range of about 10.5 to 11.5 um, or another suitable value. In further embodiments, the second epitaxial layer **606** may be or comprise a p-type epitaxial silicon layer having a thickness of about 2 um, 4 um, 6 um, 8 um, 10 um, within a range of about 4 to 11 um, or another suitable value. In yet further embodiments, the first and second epitaxial layers **604**, **606** are crystalline silicon.

Further, the deep well region **122** is disposed from a bottom surface of the second epitaxial layer **606** to a top surface of the second epitaxial layer **606**. The first doped regions **128** are disposed within the first epitaxial layer **604** and underlie the deep well region **122**. Further, the second doped region **120** extends from a sidewall of the STI structure **602** to a top surface of the second epitaxial layer **606**. In some embodiments, the deep well region **122** and the second doped region **120** comprise the second dopant (e.g., gallium) having the second doping type (e.g., p-type). Furthermore, the isolation structure **206** extends from the front-side surface **104f** of the semiconductor substrate **104**, through the STI structure **602**, to the back-side surface **104b** of the semiconductor substrate **104**. In some embodiments, regions **602a** of the STI structure **602** disposed between the

isolation structure **206** and corresponding first doped regions **128** may comprise the second dopant (e.g., gallium). In such embodiments, the second doped region **120** may be formed by an annealing process and/or an oxidation process (e.g., see FIGS. 17, 25, and/or 30), such that the second dopant (e.g., gallium) is driven into the regions **602a** of the STI structure **602**. In addition, the isolation structure **206** continuously extends from the front-side surface **104f** to the back-side surface **104b** of the semiconductor substrate **104**. In some embodiments, a width of the isolation structure **206** continuously decreases from the front-side surface **104f** to the back-side surface **104b** of the semiconductor substrate **104**.

FIG. 6B illustrates a cross-sectional view of some embodiments of an image sensor **600b** corresponding to some alternative embodiments of the image sensor **600a** of FIG. 6A, in which the doped liner **304** is disposed between the isolation structure **206** and the semiconductor substrate **104**. The doped liner **304** extends continuously from a top surface of the STI structure **602** to the back-side surface **104b** of the semiconductor substrate **104**.

FIG. 6C illustrates a cross-sectional view of some embodiments of an image sensor **600c** corresponding to some alternative embodiments of the image sensor **600b** of FIG. 6B, in which a width of the passivation layer **202** and the trench fill layer **204** discretely decrease from the front-side surface **104f** of the semiconductor substrate to the back-side surface **104b**. Here, the doped liner **304** may, for example, be or comprise an epitaxial silicon layer doped with the second dopant (e.g., gallium).

FIGS. 7-11 illustrate cross-sectional views **700-1100** of some embodiments of a first method of forming an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors, according to the present disclosure. Although the cross-sectional views **700-1100** shown in FIGS. 7-11 are described with reference to the first method, it will be appreciated that the structures shown in FIGS. 7-11 are not limited to the first method but rather may stand alone separate of the first method. Furthermore, although FIGS. 7-11 are described as a series of acts, it will be appreciated that these acts are not limited in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

As illustrated in cross-sectional view **700** of FIG. 7, a plurality of photodetectors **126** are formed within a semiconductor substrate **104**. In some embodiments, the semiconductor substrate **104** may, for example, be or comprise a bulk substrate (e.g., a bulk silicon substrate), a silicon on insulator (SOI) substrate, crystalline silicon, monocrystalline silicon, epitaxial silicon, silicon germanium (SiGe), doped epitaxial silicon, an oxygen free silicon substrate, an oxygen rich silicon substrate, P-doped silicon, or another suitable semiconductor material. Further, the semiconductor substrate **104** comprises a front-side surface **104f** that is opposite a back-side surface **104b**. The plurality of photodetectors **126** respectively comprise a first doped region **128** disposed within the semiconductor substrate **104**. In various embodiments, a process for forming the plurality of photodetectors **126** includes: selectively forming a masking layer **702** over the front-side surface **104f** of the semiconductor substrate **104**; performing a selective ion implantation process according to the masking layer **702**, thereby implanting

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one or more dopants within the semiconductor substrate **104** and forming the first doped regions **128**; and performing a removal process to remove the masking layer **702** (not shown). The one or more dopants may, for example, be or comprise phosphorus, arsenic, antimony, another suitable n-type dopant, or any combination of the foregoing with a first doping type (e.g., n-type). Thus, the first doped regions **128** have the first doping type (e.g., n-type).

As illustrated in cross-sectional view **800** of FIG. **8**, a second doped region **120** and a deep well region **122** are formed within the semiconductor substrate **104**. In some embodiments, a process for forming the second doped region **120** and/or the deep well region **122** comprises: selectively forming a masking layer **802** over the front-side surface **104f**; performing a selective ion implantation process according to the masking layer **802**, thereby implanting a second dopant **804** within the semiconductor substrate **104** and forming the second doped region **120** and the deep well region **122**; and performing a removal process to remove the masking layer **802** (not shown). The second dopant **804** may, for example, be or comprise gallium (e.g., Ga³⁺) with a second doping type (e.g., p-type) and that has a low likelihood to diffuse out of the second doped region **120** and/or the deep well region **122** during subsequent processing steps and/or during operation of the photodetectors **126**. Thus, the second doped region **120** and the deep well region **122** have the second doping type (e.g., p-type) that is opposite the first doping type (e.g., n-type). In further embodiments, the second doped region **120** and the deep well region **122** have a first doping concentration of the second dopant (e.g., gallium) ranging between about 10^{14} to 2×10^{18} atoms/cm³, or another suitable value. Further, the second dopant has a sufficiently large atomic size (e.g., about 62 picometers (pm)), such that diffusion of the second dopant is mitigated and an area of the semiconductor substrate **104** the second doped region **120** and the deep well region **122** occupies is not increased. This maintains the size of the first doped regions **128** such that a volume of the photodetectors **126** is not decreased as the semiconductor substrate **104** is exposed to heat, thereby increasing and/or maintaining a full well capacity of the photodetectors **126**.

In yet further embodiments, a diffusivity of the second dopant (e.g., gallium) at relatively high temperatures (e.g., about 700 degrees Celsius or greater) is relatively low (e.g., less than about 5.4×10^{-15} cm²/s). This, in part, mitigates diffusion of the second dopant during subsequent front-end of line formation processes (e.g., see FIG. **9**) and/or during operation of the image sensor **100**. The relatively low diffusivity of the second dopant (e.g., gallium) is beneficial to high thermal budget products, especially compared to other dopants (e.g., boron) with high diffusivity at the relatively high temperatures, which have a low thermal budget resistance and are more likely to diffuse into the first doped regions **128** during subsequent processing steps. Thus, the second doped region **120** and the deep well region **122** have a high thermal budget resistance, thereby maintaining a size of the second doped region **120** and the deep well region **122** during subsequent processing steps.

As illustrated in cross-sectional view **900** of FIG. **9**, a plurality of pixel devices **112** and an interconnect structure **102** are formed along the front-side surface **104f** of the semiconductor substrate **104**. In some embodiments, each pixel device **112** comprises a gate dielectric layer **114**, a gate electrode **116**, and a sidewall spacer structure **118**. The gate dielectric layer **114** is disposed between the gate electrode **116** and the semiconductor substrate **104**. Further, the interconnect structure **102** comprises an interconnect dielectric

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structure **106**, a plurality of conductive wires **108**, and a plurality of conductive vias **110**. In some embodiments, the interconnect dielectric structure **106** may be formed by one or more deposition processes such as a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, another suitable growth or deposition process, or any combination of the foregoing. In further embodiments, the plurality of conductive wires **108** and/or the plurality of conductive vias **110** may, for example, be formed by a single damascene process, a dual damascene process, or another suitable formation process. Furthermore, as illustrated in FIG. **9**, a thinning process is performed on the semiconductor substrate **104** to reduce the semiconductor substrate **104** from an initial thickness T_i to a thickness T_s . The thinning process may, for example, include performing a chemical mechanical planarization (CMP) process, a mechanical grinding process, another suitable thinning process, or any combination of the foregoing.

As illustrated in cross-sectional view **1000** of FIG. **10**, an isolation structure **206** is formed into the back-side surface **104b** of the semiconductor substrate **104**. In some embodiments, the isolation structure **206** comprises a passivation layer **202** and a trench fill layer **204**, where the passivation layer **202** is disposed between the semiconductor substrate **104** and the trench fill layer **204**. In some embodiments, a process for forming the isolation structure **206** includes: selectively etching the back-side surface **104b** of the semiconductor substrate **104** to form an isolation structure opening with the semiconductor substrate **104**; depositing (e.g., by CVD, PVD, ALD, etc.) the passivation layer **202** over the semiconductor substrate **104**, thereby lining the isolation structure opening; depositing (e.g., by CVD, PVD, ALD, etc.) the trench fill layer **204** over the passivation layer **202**; and performing a planarization process into the passivation layer **202** and the trench fill layer **204**, thereby forming the isolation structure **206**.

As illustrated in cross-sectional view **1100** of FIG. **11**, a plurality of light filters **130** (e.g., color filters) are formed over the back-side surface **104b** of the semiconductor substrate **104** and the isolation structure **206**. Further, a plurality of micro-lenses **132** are formed over the plurality of light filters **130**. In some embodiments, the light filters **130** and the micro-lenses **132** may be deposited by, for example, CVD, PVD, ALD, or another suitable deposition or growth process.

FIGS. **12-22** illustrate cross-sectional views **1200-2200** of some embodiments of a second method of forming an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors, according to the present disclosure. Although the cross-sectional views **1200-2200** shown in FIGS. **12-22** are described with reference to the second method, it will be appreciated that the structures shown in FIGS. **12-22** are not limited to the second method but rather may stand alone separate of the second method. Furthermore, although FIGS. **12-22** are described as a series of acts, it will be appreciated that these acts are not limited in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

As illustrated in cross-sectional view **1200** of FIG. **12**, a plurality of photodetectors **126** are formed within a semiconductor substrate **104**. The plurality of photodetectors **126**

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respectively comprise a first doped region **128** disposed within the semiconductor substrate **104**, where the first doped region **128** comprises a first doping type (e.g., n-type). In some embodiments, the semiconductor substrate **104** may be or comprise any type of semiconductor body (e.g., silicon/germanium/CMOS bulk, SiGe, SOI, etc.) such as a semiconductor wafer or one or more die on a wafer, as well as any other type of semiconductor and/or epitaxial layers otherwise associated therewith. For example, the semiconductor substrate **104** may be or comprise a first epitaxial layer **604** overlying a second epitaxial layer **606**. In some embodiments, the first epitaxial layer **604** may be or comprise a p-type epitaxial silicon layer having a thickness T_f of about 5.4 micrometers (μm), about 6 μm , within a range of about 5.4 to 6 μm , or another suitable value. In yet further embodiments, the first epitaxial layer **604** may be or comprise an n-type epitaxial silicon layer having a thickness T_f of about 11 μm , within a range of about 10.5 to 11.5 μm , or another suitable value. In further embodiments, the second epitaxial layer **606** may be or comprise a p-type epitaxial silicon layer having an initial thickness T_{ie} within a range of about 4 to 11 μm , or another suitable value. The first epitaxial layer **604** may, for example, be formed by p-type epitaxial process, an n-type epitaxial process, or another suitable epitaxial process. In yet further embodiments, the first and second epitaxial layers **604**, **606** are crystalline silicon.

In some embodiments, the first epitaxial layer **604** and/or the second epitaxial layer **606** each comprise a second dopant (e.g., gallium) having a second doping type (e.g., p-type) opposite the first doping type (e.g., n-type). In some embodiments, a doping concentration of the first epitaxial layer **604** may, for example, be within a range of about 10^{14} to 5×10^{15} atoms/ cm^3 , or another suitable value. Further, a deep well region **122** is formed within the second epitaxial layer **606**, the deep well region **122** may be formed by a selective ion implantation process, or another suitable process. In some embodiments, the deep well region **122** comprises the second dopant (e.g., gallium) with a doping concentration within a range of about 10^{14} to 2×10^{18} atoms/ cm^3 , or another suitable value. In various embodiments, a process for forming the plurality of photodetectors **126** includes: selectively forming a masking layer (not shown) over a front-side surface **104f** of the semiconductor substrate **104**; performing a selective implantation process according to the masking layer, thereby implanting one or more dopants within the semiconductor substrate **104** and forming the first doped regions **128**; and performing a removal process to remove the masking layer (not shown). The one or more dopants may, for example, be or comprise phosphorus, arsenic, antimony, another suitable n-type dopant, or any combination of the foregoing with the first doping type (e.g., n-type).

As illustrated in cross-sectional view **1300** of FIG. **13**, a masking layer **1302** is formed over the front-side surface **104f** of the semiconductor substrate **104**, a shallow trench isolation (STI) structure **602** is formed within the semiconductor substrate **104** and the masking layer **1302**, and a photoresist structure **1310** is formed over the masking layer **1302**. In some embodiments, the masking layer **1302** comprises a first dielectric layer **1304**, a second dielectric layer **1306**, and a third dielectric layer **1308**. In some embodiments, the masking layer **1302** is formed by multiple deposition processes such as one or more PVD processes, CVD processes, ALD processes, thermal oxidation processes, other suitable deposition or growth processes, or any combination of the foregoing. The first dielectric layer **1304** may,

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for example, be or comprise silicon dioxide, another suitable dielectric material and may be formed to a thickness of about 90 angstroms, or another suitable thickness value. The second dielectric layer **1306** may, for example, be or comprise silicon nitride, or another suitable dielectric material and may be formed to a thickness of about 800 angstroms, or another suitable thickness value. The third dielectric layer **1308** may, for example, be or comprise a layer of silicon dioxide, or another suitable dielectric material formed to a thickness of about 270 angstroms and a layer of silicon oxynitride, or another suitable dielectric material formed to a thickness of about 3,000 angstroms. It will be appreciated that layers of the third dielectric layer **1308** having other thickness values is within the scope of the disclosure.

In some embodiments, a process for forming the STI structure **602** may comprise: selectively etching the semiconductor substrate **104**, the first dielectric layer **1304**, and the second dielectric layer **1306** to form an STI opening; depositing (e.g., by CVD, PVD, ALD, etc.) a dielectric material (e.g., silicon dioxide, silicon nitride, silicon carbide, silicon oxynitride, silicon oxycarbide, etc.) within the STI opening; and performing a planarization process (e.g., a CMP process) and/or an etch back process on the dielectric material, thereby forming the STI structure **602**. Further, in some embodiments, the photoresist structure **1310** comprises a photoresist formed to a thickness of about 6900 angstroms and an anti-reflective layer formed to a thickness of about 800 angstroms. It will be appreciated that layers of the photoresist structure **1310** having other thickness values is within the scope of the disclosure. The photoresist structure **1310** may, for example, be deposited by a PVD process, a CVD process, an ALD process, a spin-on process, another suitable growth or deposition process, or any combination of the foregoing.

As illustrated in cross-sectional view **1400** of FIG. **14**, a patterning process is performed on the masking layer **1302**, the STI structure **602**, and the semiconductor substrate **104**, thereby forming an isolation structure opening **1402** within the semiconductor substrate **104**. In some embodiments, the patterning process may include performing a wet etch process, a dry etch process, another suitable etch process, or any combination of the foregoing. The isolation structure opening **1402** extends a distance d_1 into the front-side surface **104f** of the semiconductor substrate **104** and has a width w_1 . In some embodiments, the distance d_1 is about 7 μm , 8 μm , 9 μm , 10 μm , within a range of about 7 to 10 μm , or another suitable value. In further embodiments, the width w_1 is about 0.3 μm , within a range of about 0.25 to 0.35 μm , or another suitable value.

As illustrated in cross-sectional view **1500** of FIG. **15**, a doped liner **304** is formed along sidewalls and a lower surface of the semiconductor substrate **104** that define the isolation structure opening **1402**. In yet further embodiments, the doped liner **304** may be referred to as a doped region of the semiconductor substrate **104**. In some embodiments, a process for forming the doped liner **304** may comprise performing a plasma doping process according to the photoresist structure **1310**, thereby doping sidewalls and a lower surface of the semiconductor substrate **104** with the second dopant (e.g., gallium). In yet further embodiments, the plasma doping process is performed such that crystalline regions of the semiconductor substrate **104** are amorphized, thus the doped liner **304** comprises amorphous silicon doped with the second dopant (e.g., gallium). The second dopant may, for example, be or comprise gallium having the second doping type (e.g., p-type). In some embodiments, the doped liner **304** comprises the second dopant and has a doping

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concentration of the second dopant ranging between about 10^{14} to 2×10^{18} atoms/cm³, 5×10^{15} to 2×10^{18} atoms/cm³ or another suitable value. In various embodiments, the doping concentration of the second dopant of the doped liner 304 adjacent the first epitaxial layer 604 might be more than the concentration of the second dopant of the doped liner 304 adjacent the second epitaxial layer 606, due to the second dopant originally existing in the first epitaxial layer 604. Accordingly, the doping concentration of the second dopant of the upper doped liner 304 might be more than that of the adjacent first epitaxial layer 604 (more than 10^{14} atoms/cm³), while the concentration of the second dopant of the lower doped liner 304 adjacent the second epitaxial layer 606 is about 10^{14} atoms/cm³. In yet further embodiments, a thickness of the doped liner 304 is greater than about 10 nanometers (nm) or another suitable thickness value. In yet further embodiments, the doped liner 304 extends along sidewalls of the STI structure 602, sidewalls of the mask layer 1302, and/or sidewalls of the photoresist structure 1310 that define the isolation structure opening 1402 (not shown).

As illustrated in cross-sectional view 1600 of FIG. 16, an oxidation process is performed on the doped liner 304, such that a doped dielectric layer 1602 is formed along sidewalls and the lower surface of the semiconductor substrate 104 that define the isolation structure opening 1402. In some embodiments, the doped dielectric layer 1602 comprises silicon dioxide doped with the second dopant (e.g., gallium). In yet further embodiments, at least a portion of the doped liner 304 is converted to silicon dioxide doped with the second dopant by the oxidation process. In some embodiments, the oxidation process includes performing an in-situ steam generation (ISSG) process or another suitable oxidation process. In yet further embodiments, the oxidation process may cure defects (e.g., crystalline defects) along the sidewalls and the lower surface of the semiconductor substrate 104 as a result of the plasma doping process of FIG. 15.

As illustrated in cross-sectional view 1700 of FIG. 17, an annealing process is performed on the semiconductor substrate 104, thereby forming a second doped region 120 within the semiconductor substrate 104. In some embodiments, the second doped region 120 comprises the second dopant (e.g., gallium) and has a doping concentration within a range of about 10^{14} to 2×10^{18} atoms/cm³. In yet further embodiments, the annealing process may drive the second dopant (e.g., gallium) from the doped liner 304 and/or the doped dielectric layer 1602 into the semiconductor substrate 104 to form the second doped region 120, such that the second doped region 120 is self-aligned along the isolation structure opening 1402. In various embodiments, the annealing process drives the second dopant (e.g., gallium) a distance of at least 10 nm from the doped liner 304 into the semiconductor substrate 104. It will be appreciated that the annealing process driving the second dopant other distance values into the semiconductor substrate 104 is within the scope of the disclosure. The annealing process is performed by exposing the semiconductor substrate 104 to an elevated temperature that is, for example, within a range of about 500 to 950 degrees Celsius, or another suitable value. Further, the annealing process may cure crystalline defects and/or activate the second dopant (e.g. gallium) within the semiconductor substrate 104. In yet further embodiments, the annealing process may convert the amorphous silicon within the doped liner 304 into crystalline silicon, such that the doped liner 304 and the second doped region 120 are one and the same.

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As illustrated in cross-sectional view 1800 of FIG. 18, a removal process is performed to remove the doped dielectric layer (1602 of FIG. 17). In some embodiments, the removal process includes performing a wet etch process and exposing the semiconductor substrate 104 to a wet etchant such as hydrofluoric acid. In various embodiments, the removal process is configured to remove the doped dielectric layer (1602 of FIG. 17) (e.g., silicon dioxide) and may remove at least a portion of the doped liner (304 of FIG. 17) that comprises silicon dioxide. In some embodiments, the doped liner 304 might remain, and the doped liner 304 and the second doped region 120 are substantially the same material so the doped liner 304 is not additionally marked. Further, a planarization process (e.g., a CMP process) and/or an etch back process is performed on the semiconductor substrate 104 to remove the photoresist structure (1310 of FIG. 17) and/or the masking layer (1302 of FIG. 17).

As illustrated in cross-sectional view 1900 of FIG. 19, a passivation film 1902 is deposited over the front-side surface 104f of the semiconductor substrate 104 and lines the isolation structure opening (1402 of FIG. 18). Further, a trench fill structure 1904 is deposited over the passivation film 1902 and fills a remaining portion of the isolation structure opening (1402 of FIG. 18). In some embodiments, the passivation film 1902 and/or the trench fill structure 1904 are respectively deposited by CVD, PVD, ALD, or another suitable deposition or growth process. In some embodiments, the passivation film 1902 may, for example, be or comprise aluminum oxide (Al₂O₃), hafnium oxide (HfO₂), titanium oxide (Ta₂O₅), another high-k dielectric material, any combination of the foregoing, or the like. The trench fill structure 1904 may, for example, be or comprise silicon dioxide, silicon nitride, silicon carbide, another dielectric material, or any combination of the foregoing.

As illustrated in cross-sectional view 2000 of FIG. 20, a planarization process (e.g., a CMP process) is performed into the passivation film (1902 of FIG. 19) and the trench fill structure (1904 of FIG. 19), thereby forming a passivation layer 202 and a trench fill layer 204, respectively. Further, the planarization process forms an isolation structure 206 that extends from the front-side surface 104f of the semiconductor substrate 104 to a point below the front-side surface 104f. In some embodiments, the isolation structure 206 is disposed between adjacent photodetectors 126 and comprises the passivation layer 202, the trench fill layer 204, and the STI structure 602.

As illustrated in cross-sectional view 2100 of FIG. 21, a plurality of pixel devices 112 and an interconnect structure 102 are formed along the front-side surface 104f of the semiconductor substrate 104. In some embodiments, the interconnect structure 102 is formed as illustrated and/or described in FIG. 9. Further, as illustrated in FIG. 21, a thinning process is performed on the semiconductor substrate 104 to reduce the semiconductor substrate 104 from an initial thickness Ti to a thickness Ts. The thinning process may, for example, include performing a CMP process, a mechanical grinding process, another suitable thinning process, or any combination of the foregoing. In some embodiments, the thinning process removes at least a portion of the isolation structure 206 and/or at least a portion of the second epitaxial layer 606.

As illustrated in cross-sectional view 2200 of FIG. 22, a plurality of light filters 130 (e.g., color filters) are formed over the back-side surface 104b of the semiconductor substrate 104 and the isolation structure 206. Further, a plurality of micro-lenses 132 are formed over the plurality of light filters 130. In some embodiments, the light filters 130 and

the micro-lenses **132** may be deposited by, for example, CVD, PVD, ALD, or another suitable deposition or growth process.

FIGS. **23-26** illustrate cross-sectional views **2300-2600** of a second embodiment of the second method of FIGS. **12-22**. For example, FIGS. **23-26** illustrates cross-sectional views **2300-2600** of some alternative embodiments of acts that may be performed in place of the acts at FIGS. **15-18**, such that the second method of FIGS. **12-22** may alternatively proceed from FIGS. **12-14** to FIG. **23**, FIG. **23** to FIGS. **24-26**, and then from FIG. **26** to FIGS. **19-22** (i.e., skipping FIGS. **15-18**).

As illustrated in cross-sectional view **2300** of FIG. **23**, a dielectric layer **2302** is formed along sidewalls and the lower surface of the semiconductor substrate **104**. In some embodiments, the dielectric layer **2302** is formed by CVD, PVD, thermal oxidation, in-situ steam generation (ISSG), or another suitable deposition or growth process. In some embodiments, the dielectric layer **2302** is formed by an oxidation process (e.g., such as an ISSG process), that may cure defects (e.g., crystalline defects) along the sidewalls and the lower surface of the semiconductor substrate **104** as a result of the patterning process of FIG. **14**. In various embodiments, the dielectric layer **2302** may be or comprise an oxide, such as silicon dioxide, another suitable dielectric material, or the like. In yet further embodiments, the dielectric layer **2302** may be referred to as a protection layer that is configured to mitigate damage to the semiconductor substrate **104** during subsequent processing steps (e.g., the plasma doping process of FIG. **24**).

As illustrated in cross-sectional view **2400** of FIG. **24**, a doped liner **304** is formed under the dielectric layer **2302**. The doped liner **304** may be formed by a plasma doping process. In yet further embodiments, the doped liner **304** comprises the second dopant (e.g., gallium) and is formed to a thickness greater than 10 nm, or another suitable thickness value. In further embodiments, after the plasma doping process, the dielectric layer **2302** may comprise the second dopant (e.g., gallium). The second dopant may, for example, be or comprise gallium having the second doping type (e.g., p-type). In some embodiments, the doped liner **304** has a doping concentration of the second dopant ranging between about 10^{14} to 2×10^{18} atoms/cm³ or another suitable value. In yet further embodiments, the doped liner **304** may be or comprise amorphous silicon doped with the second dopant.

As illustrated in cross-sectional view **2500** of FIG. **25**, an annealing process is performed on the semiconductor substrate **104**, thereby forming a second doped region **120** within the semiconductor substrate **104**. In some embodiments, the second doped region **120** comprises the second dopant (e.g., gallium) and has a doping concentration within a range of about 10^{14} to 2×10^{18} atoms/cm³, or another suitable value. In yet further embodiments, the annealing process drives the second dopant (e.g., gallium) from the doped liner (**304** of FIG. **24**) and/or the dielectric layer **2302** into the semiconductor substrate **104** to form the second doped region **120**, such that the second doped region **120** is self-aligned along the isolation structure opening **1402**. In various embodiments, the annealing process drives the second dopant (e.g., gallium) a distance of at least 10 nm from the doped liner **304** into the semiconductor substrate **104**. It will be appreciated that the annealing process driving the second dopant other distance values into the semiconductor substrate **104** is within the scope of the disclosure. The annealing process is performed by exposing the semiconductor substrate **104** to an elevated temperature that is, for example, within a range of about 500 to 950 degrees Celsius,

or another suitable value. Further, the annealing process may cure crystalline defects and/or activate the second dopant (e.g. gallium) within the semiconductor substrate **104**. In yet further embodiments, the annealing process may convert the amorphous silicon within the doped liner **304** into crystalline silicon, such that the doped liner **304** and the second doped region **120** are one and the same.

As illustrated in cross-sectional view **2600** of FIG. **26**, a removal process is performed to remove the dielectric layer (**2302** of FIG. **25**). In some embodiments, the removal process includes performing a wet etch process and exposing the semiconductor substrate **104** to a wet etchant such as hydrofluoric acid. Further, a planarization process (e.g., a CMP process) and/or an etch back process is performed on the semiconductor substrate **104** to remove the photoresist structure (**1310** of FIG. **25**) and/or the masking layer (**1302** of FIG. **25**).

FIGS. **27-31** illustrate cross-sectional views **2700-3100** of a third embodiment of the second method of FIGS. **12-22**. For example, FIGS. **27-31** illustrates cross-sectional views **2700-3100** of some alternative embodiments of acts that may be performed in place of the acts at FIGS. **15-20**, such that the second method of FIGS. **12-22** may alternatively proceed from FIGS. **12-14** to FIG. **27**, FIG. **27** to FIGS. **28-31**, and then from FIG. **31** to FIGS. **21-22** (i.e., skipping FIGS. **15-20**).

As illustrated in cross-sectional view **2700** of FIG. **27**, a dielectric layer **2702** is formed along sidewalls and the lower surface of the semiconductor substrate **104**. In some embodiments, the dielectric layer **2702** is formed by CVD, PVD, thermal oxidation, in-situ steam generation (ISSG), or another suitable deposition or growth process. In some embodiments, the dielectric layer **2702** is formed by an oxidation process (e.g., such as an ISSG process), that may cure defects (e.g., crystalline defects) along the sidewalls and the lower surface of the semiconductor substrate **104** as a result of the patterning process of FIG. **14**. In various embodiments, the dielectric layer **2702** may be or comprise an oxide, such as silicon dioxide, another suitable dielectric material, or the like.

As illustrated in cross-sectional view **2800** of FIG. **28**, a removal process is performed to remove the dielectric layer (**2702** of FIG. **27**). In some embodiments, the removal process includes performing a wet etch process and exposing the semiconductor substrate **104** to a wet etchant such as hydrofluoric acid.

As illustrated in cross-sectional view **2900** of FIG. **29**, a doped liner **304** is formed along sidewalls and a lower surface of the semiconductor substrate **104** that define the isolation structure opening **1402**. In some embodiments, the doped liner **304** is formed by a selective epitaxial growth process. The selective epitaxial growth process may, for example, be an epitaxial process or another form of a deposition process such as CVD, plasma enhanced chemical vapor deposition (PE-CVD), ALD, PVD, or another suitable growth or deposition process. In further embodiments, the doped liner **304** is selectively grown such that the doped liner **304** is in-situ doped with the second dopant (e.g., gallium). The doped liner **304** may, for example, be or comprise epitaxial silicon doped with the second dopant (e.g., gallium) and has a doping concentration of the second dopant within a range of about 10^{14} to 2×10^{18} atoms/cm³ or another suitable value.

As illustrated in cross-sectional view **3000** of FIG. **30**, an oxidation process is performed on the doped liner **304**, thereby forming a dielectric layer **3002** over the doped liner **304** and a second doped region **120** within the semiconduc-

tor substrate **104**. In some embodiments, the dielectric layer **3002** may be or comprise an oxide such as silicon dioxide and the second dopant (e.g., gallium). In yet further embodiments, the oxidation process converts at least a portion of the doped liner **304** to silicon dioxide doped with the second dopant (e.g., gallium). In some embodiments, the oxidation process includes performing an in-situ steam generation (ISSG) process or another suitable oxidation process. In yet further embodiments, the oxidation process includes exposing the semiconductor substrate **104** to an elevated temperature that is, for example, within a range of about 500 to 950 degrees Celsius, or another suitable value. In such embodiments, the oxidation process drives the second dopant (e.g., gallium) from the doped liner **304** and/or the dielectric layer **3002** into the semiconductor substrate **104**, thereby forming the second doped region **120** within the semiconductor substrate **104**. In various embodiments, the oxidation process drives the second dopant (e.g., gallium) a distance of at least 10 nm from the doped liner **304** into the semiconductor substrate **104**. It will be appreciated that the annealing process driving the second dopant other distance values into the semiconductor substrate **104** is within the scope of the disclosure. Thus, the second doped region **120** is self-aligned along sidewalls and a lower surface of the semiconductor substrate **104** that defines the isolation structure opening **1402**. In some embodiments, the second doped region **120** comprises the second dopant (e.g., gallium) and has a doping concentration within a range of about 10^{14} to 2×10^{18} atoms/cm³ or another suitable value. In yet further embodiments, after forming the second doped region **120**, a removal process is performed to remove the dielectric layer **3002** and/or the doped liner **304** (not shown).

As illustrated in cross-sectional view **3100** of FIG. **31**, an isolation structure **206** is formed within the semiconductor substrate **104**. In some embodiments, the isolation structure **206** comprises a passivation layer **202** and a trench fill layer **204**. In further embodiments, a process for forming the isolation structure **206** may include depositing (e.g., by CVD, PVD, ALD, etc.) the passivation layer **202** within the isolation structure opening (**1402** of FIG. **30**); depositing (e.g., by CVD, PVD, ALD, etc.) the trench fill layer **204** over the passivation layer **202**, thereby filling the isolation structure opening (**1402** of FIG. **30**); and performing a planarization process (e.g., a CMP process) and/or an etch back process to remove at least a portion of the passivation layer **202** and/or the trench fill layer **204**. In such embodiments, the planarization process and/or the etch back process can remove the photoresist structure (**1310** of FIG. **30**) and/or the masking layer (**1302** of FIG. **30**) from over the front-side surface **104f** of the semiconductor substrate **104**.

FIG. **32** illustrates a method **3200** of forming an image sensor comprising a plurality of photodetectors having first doped regions and a second doped region laterally surrounding the first doped regions, where the second doped region comprises a second dopant configured to improve full well capacity of the photodetectors in accordance with some embodiments. Although the method **3200** is illustrated and/or described as a series of acts or events, it will be appreciated that the method is not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included.

At act **3202**, a first doped region is formed within a semiconductor substrate. The first doped region comprises a first doping type. FIG. **7** illustrates a cross-sectional view **700** corresponding to some embodiments of act **3202**. FIG. **12** illustrates a cross-sectional view **1200** corresponding to some alternative embodiments of act **3202**.

At act **3204**, a deep well region is formed within the semiconductor substrate. The deep well region comprises a second doping type opposite the first doping type. FIG. **8** illustrates a cross-sectional view **800** corresponding to some embodiments of act **3204**. FIG. **12** illustrates a cross-sectional view **1200** corresponding to some alternative embodiments of act **3204**.

At act **3206**, a second doped region is formed within the semiconductor substrate. The second doped region abuts the first doped region and comprises a second dopant having the second doping type. The second dopant comprises gallium. In some embodiments, the deep well region comprises the second dopant. FIG. **8** illustrates a cross-sectional view **800** corresponding to some embodiments of act **3206**. FIGS. **15-18** illustrate cross-sectional views **1500-1800** corresponding to some alternative embodiments of act **3206**. Further, FIGS. **23-26** illustrate cross-sectional views **2300-2600** corresponding to some further alternative embodiments of act **3206**. Furthermore, FIGS. **27-30** illustrate cross-sectional views **2700-3000** corresponding to yet another alternative embodiment of act **3206**.

At act **3208**, an interconnect structure and a plurality of pixel devices are formed over a front-side surface of the semiconductor substrate. FIG. **9** illustrates a cross-sectional view **900** corresponding to some embodiments of act **3208**. FIG. **21** illustrates a cross-sectional view **2100** corresponding to some alternative embodiments of act **3208**.

At act **3210**, an isolation structure is formed within the semiconductor substrate such that the isolation structure laterally surrounds the first doped region. FIG. **10** illustrates a cross-sectional view **1000** corresponding to some embodiments of act **3210**. FIGS. **14**, **19**, and **20** illustrate cross-sectional views **1400**, **1900**, and **2000** corresponding to some alternative embodiments of act **3210**.

At act **3212**, a plurality of light filters are formed over a back-side surface of the semiconductor substrate and a plurality of micro-lenses are formed over the plurality of light filters. FIG. **11** illustrates a cross-sectional view **1100** corresponding to some embodiments of act **3212**. FIG. **22** illustrates a cross-sectional view **2200** corresponding to some alternative embodiments of act **3212**.

Accordingly, in some embodiments, the present disclosure relates to an image sensor comprising a first doped region disposed within a semiconductor substrate, where the first doped region comprises a first dopant having a first doping type. A second doped region abuts the first doped region and comprises a second dopant having a second doping type opposite the first doping type, where the second dopant comprises gallium.

In some embodiments, the present application provides an image sensor including a photodetector disposed in a semiconductor substrate, wherein the photodetector comprises a first doped region comprising a first dopant having a first doping type; a deep well region disposed within the semiconductor substrate, wherein the deep well region extends from a back-side surface of the semiconductor substrate to a top surface of the first doped region; and a second doped region disposed within the semiconductor substrate and abutting the first doped region, wherein the second doped region and the deep well region comprise a second dopant

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having a second doping type opposite the first doping type, wherein the second dopant comprises gallium.

In some embodiments, the present application provides an image sensor including a first photodetector and a second photodetector disposed in a semiconductor substrate, wherein the first and second photodetectors respectively comprise a first doped region having a first doping type; a plurality of pixel devices disposed along a front-side surface of the semiconductor substrate; an isolation structure extending into the semiconductor substrate, wherein the isolation structure laterally surrounds the first photodetector and the second photodetector, wherein the isolation structure comprises a trench fill layer and a passivation layer disposed between the semiconductor substrate and the trench fill layer; a deep well region disposed within the semiconductor substrate and overlying the first and second photodetectors, wherein the deep well region comprises a second doping type opposite the first doping type; and a second doped region disposed within the semiconductor substrate and extending along sidewalls of the isolation structure, wherein the second doped region comprises the second doping type and abuts the first doped region of the first and second photodetectors, wherein the second doped region is disposed between the isolation structure and the deep well region, and wherein the second doping type comprises gallium.

In some embodiments, the present application provides a method for forming an image sensor, the method includes performing a first ion implantation process on a front-side surface of a semiconductor substrate to form first doped regions within the semiconductor substrate, wherein the first doped regions comprise a first dopant having a first doping type; forming a second doped region within the semiconductor substrate such that the second doped region abuts the first doped regions, wherein the second doped region comprises a second dopant having a second doping type opposite the first doping type, wherein the second dopant comprises gallium; and forming a plurality of semiconductor devices along the front-side surface of the semiconductor substrate.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An image sensor comprising:

- a photodetector disposed in a semiconductor substrate, wherein the photodetector comprises a first doped region comprising a first dopant having a first doping type;
- a deep well region disposed within the semiconductor substrate, wherein the deep well region extends from a back-side surface of the semiconductor substrate to a top of the first doped region;
- a second doped region disposed within the semiconductor substrate and abutting the first doped region, wherein the second doped region and the deep well region comprise a second dopant having a second doping type opposite the first doping type;

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an isolation structure disposed within the semiconductor substrate, wherein the isolation structure extends from the back-side surface of the semiconductor substrate to a point below the back-side surface; and

a doped liner disposed between the isolation structure and the second doped region, wherein the doped liner comprises the second dopant, wherein the first doped region is arranged between opposing sidewalls of the doped liner, wherein the second doped region vertically extends along the opposing sidewalls of the doped liner and laterally spaces the first doped region from the opposing sidewalls of the doped liner and opposing sidewalls of the isolation structure.

2. The image sensor of claim 1, wherein a doping concentration of the first dopant within the second doped region and the deep well region is within a range of about 10^{14} to 2×10^{18} atoms/cm³.

3. The image sensor of claim 1, wherein a bottom of the second doped region is below a bottom surface of the doped liner.

4. The image sensor of claim 1, wherein the isolation structure comprises a passivation layer and a trench fill layer, wherein the passivation layer is disposed between the semiconductor substrate and the trench fill layer.

5. The image sensor of claim 1, wherein the second doped region extends continuously from the top of the first doped region, along opposing sides of the first doped region, to a bottom of the first doped region, wherein the second doped region meets the opposing sides of the first doped region at vertically extending PN junctions.

6. The image sensor of claim 1, wherein the second doped region extends continuously from a bottom of the first doped region to the back-side surface of the semiconductor substrate.

7. The image sensor of claim 1, wherein a doping concentration of the first doped region is less than a doping concentration of the second doped region.

8. An image sensor comprising:

- a first photodetector and a second photodetector disposed in a semiconductor substrate, wherein the first and second photodetectors respectively comprise a first doped region comprising a first doping type;
- a plurality of pixel devices disposed along a front-side surface of the semiconductor substrate;
- a shallow trench isolation (STI) structure disposed within the semiconductor substrate and extending from the front-side surface of the semiconductor substrate to a point above the front-side surface;
- an isolation structure extending into the semiconductor substrate, wherein the isolation structure laterally surrounds the first photodetector and the second photodetector, wherein the isolation structure comprises a trench fill layer and a passivation layer disposed between the semiconductor substrate and the trench fill layer, wherein the isolation structure extends through the STI structure; and
- a second doped region disposed within the semiconductor substrate and extending along sidewalls of the isolation structure, wherein the second doped region comprises a second doping type opposite the first doping type and abuts the first doped region of the first and second photodetectors.

9. The image sensor of claim 8, wherein the isolation structure continuously extends from the front-side surface of the semiconductor substrate to a back-side surface of the semiconductor substrate.

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10. The image sensor of claim 8, further comprising:
a doped liner disposed between the passivation layer and
the semiconductor substrate, wherein the doped liner
comprises the second doping type.
11. The image sensor of claim 10, wherein the doped liner 5
comprises epitaxial silicon.
12. The image sensor of claim 8, wherein the second
doped region extends from a sidewall of the STI structure to
a top surface of the STI structure.
13. The image sensor of claim 8, further comprising: 10
a deep well region disposed within the semiconductor
substrate and continuously extending from a back-side
surface of the semiconductor substrate to a top of the
first doped region, wherein the deep well region com-
prises the second doping type.
14. The image sensor of claim 13, wherein the second 15
doped region is disposed between the isolation structure and
the deep well region.
15. An integrated chip comprising:
a photodetector comprising a first doped region disposed 20
within a substrate, wherein the first doped region com-
prises a first doping type;
a first isolation structure disposed on a first surface of the
substrate and adjacent to the first doped region;
a second isolation structure extending through the first 25
isolation structure and extending from a top surface of
the first isolation structure towards a second surface of
the substrate opposite the first surface of the substrate;
and

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- a second doped region disposed in the substrate around
the first doped region, wherein the second doped region
extends along sidewalls of the first and second isolation
structures, wherein the second doped region comprises
a second doping type different from the first doping
type.
16. The integrated chip of claim 15, further comprising:
a doped liner disposed between the substrate and the
second isolation structure, wherein the doped liner
comprises the second doping type.
17. The integrated chip of claim 16, wherein a thickness
of the first isolation structure is greater than a thickness of
the doped liner.
18. The integrated chip of claim 15, further comprising:
a deep well region disposed between the second surface of
the substrate and the first doped region, wherein the
second doped region is disposed on opposing sides of
the deep well region, wherein the deep well region
comprises the second doping type.
19. The integrated chip of claim 15, further comprising:
a light filter disposed on the second surface of the sub-
strate; and
a micro-lens disposed on the light filter.
20. The integrated chip of claim 15, wherein a height of
the second isolation structure above the top surface of the
first isolation structure is greater than a height of the first
isolation structure.

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